TE 357: Optical Communications

Lecture 3: Optical Fibers

Lecture Objectives

1. To gain insight into the propagation of light along an optical fiber
2. To explain some phenomena that light goes through

Lecture Objective 1

- To gain insight into the propagation of light along an optical fiber
- Thus
  - To determine
    - the conditions needed to transmit light through an optical fiber
      - and resolve how these conditions can be implemented in a practical manner

Lecture Objective 2

- To explain some phenomena that light goes through
- Thus
    - Attenuation
        - Definition
        - Mechanisms
        - Loss calculation

Lecture Objective 2

- Thus
  - Dispersion
    - definition
    - Concept of modes
    - Calculation of number of modes
    - Modal (intermodal) dispersion
      - Pulse spread calculation

Introduction

- What is an optical fiber?

Introduction

- What is an optical fiber?
  - It is a thin, transparent, flexible strand
  - It is a dielectric waveguide that operates at optical frequencies

Introduction

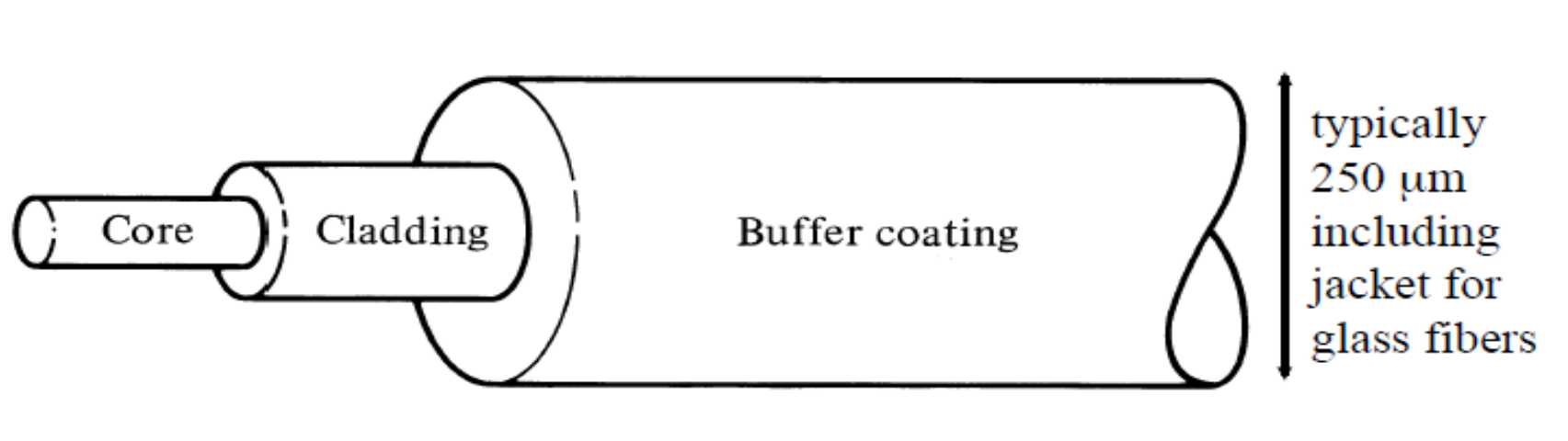
- What is an optical fiber?
  - It is normally cylindrical in form and confines electromagnetic energy in the form of light to within its surfaces.
    - It guides the light in a direction parallel to its axis

Introduction

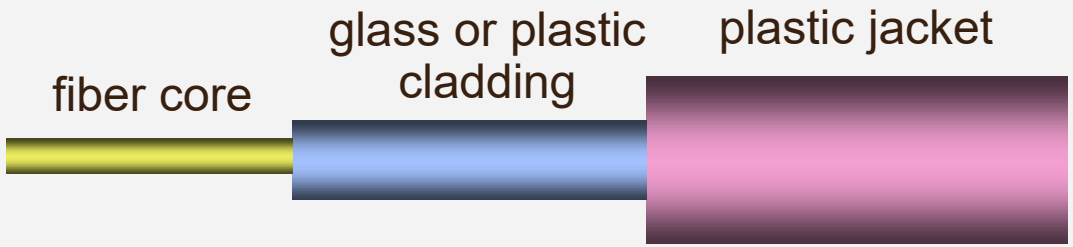
- Some questions that arise concerning optical fibers are:
  1. What is the structure of an optical fiber?
  2. Of what materials are fibers made?
  3. How does light propagate along a fiber?
  4. What is the signal loss or attenuation mechanism in a fiber?
  5. Why and to what degree does a signal get distorted as it travels along a fiber?

Optical Fiber- Basic structure

- Optical fiber structure
  - Core and cladding made from the same material (e.g. silica)
    - differ only in their refractive indexes (why?)

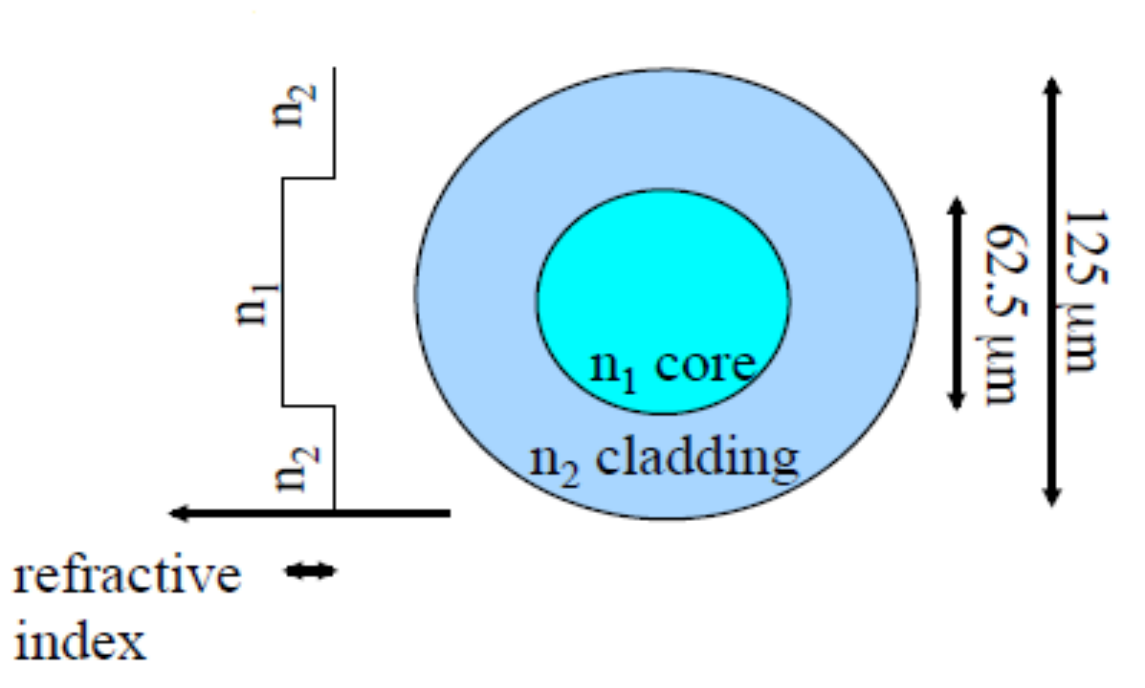


Optical Fiber-Basic structure



Optical Fiber

- Optical fiber structure



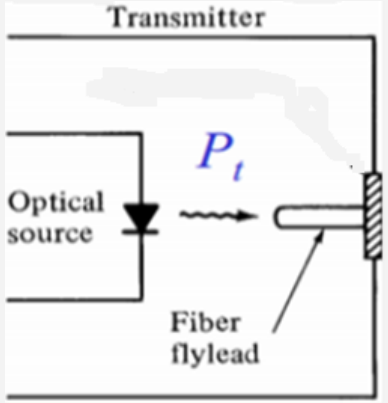
- Step index fiber

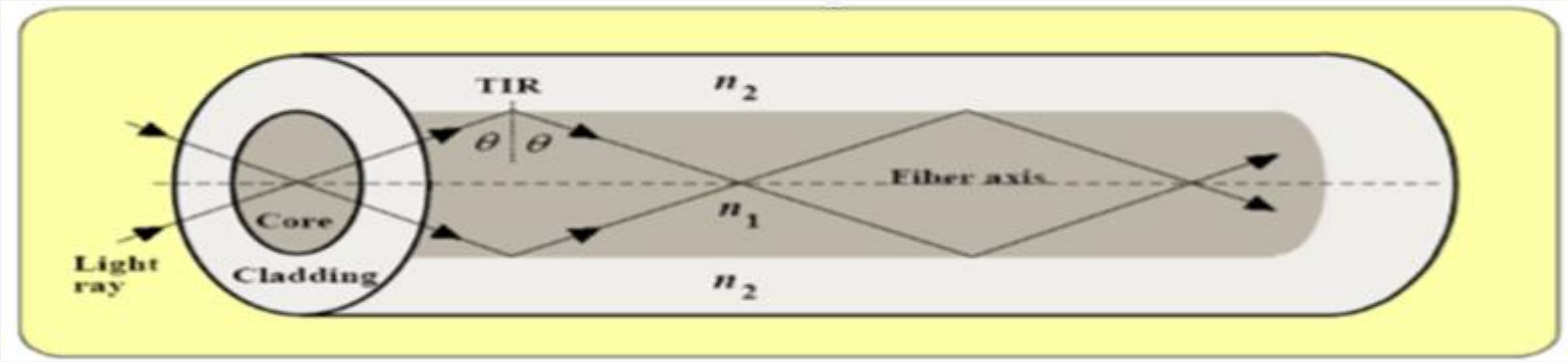
Light ray guiding condition

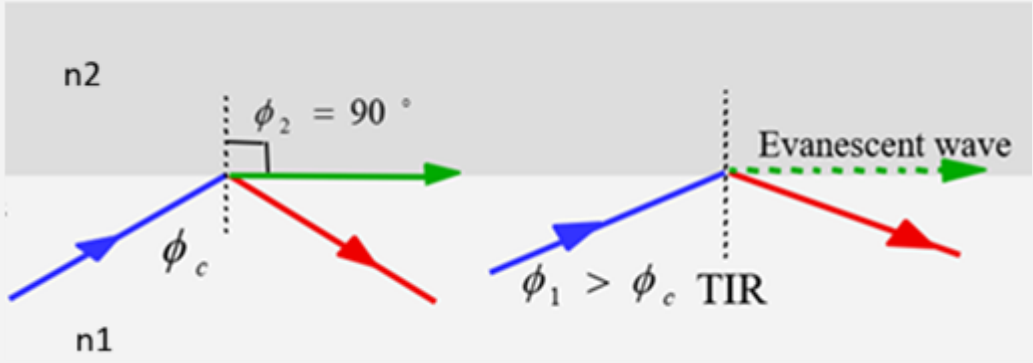
- To save light
  - inside a strand of fiber, what needs to happen?

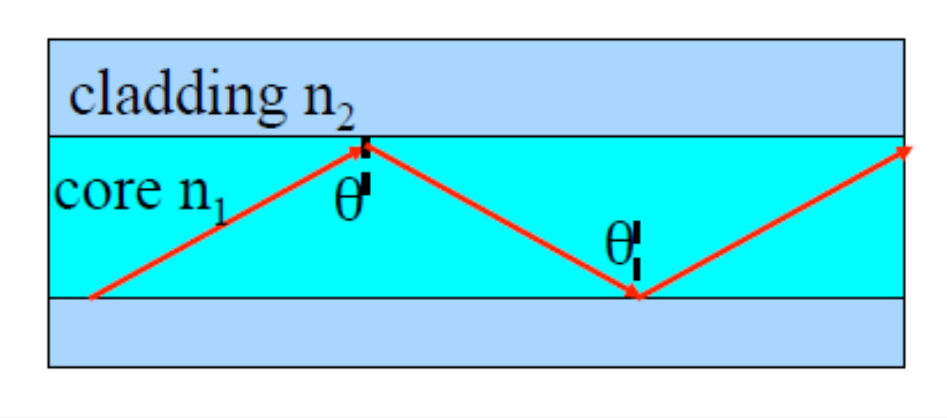
Light ray guiding condition

- Framework



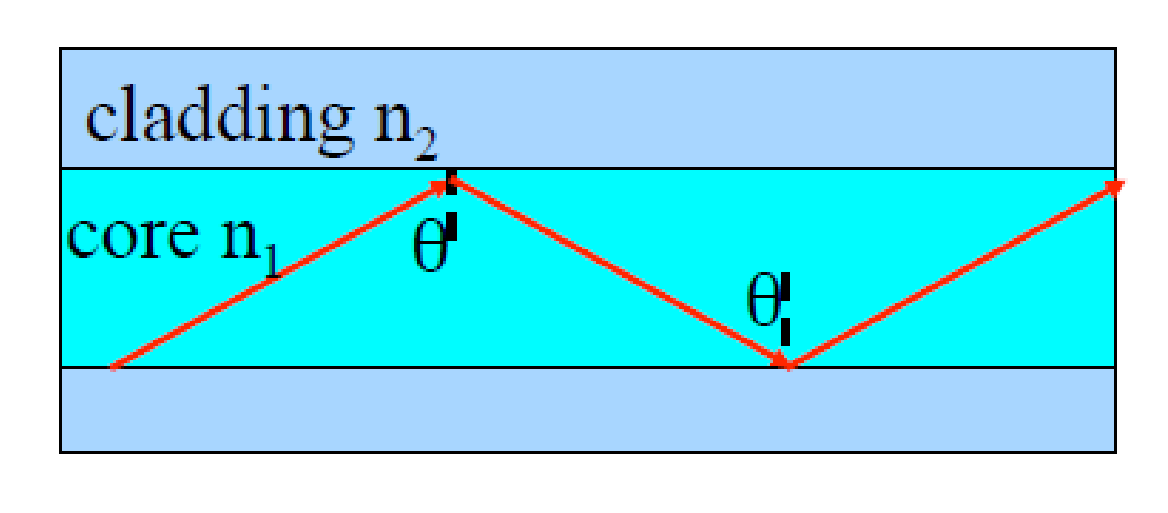






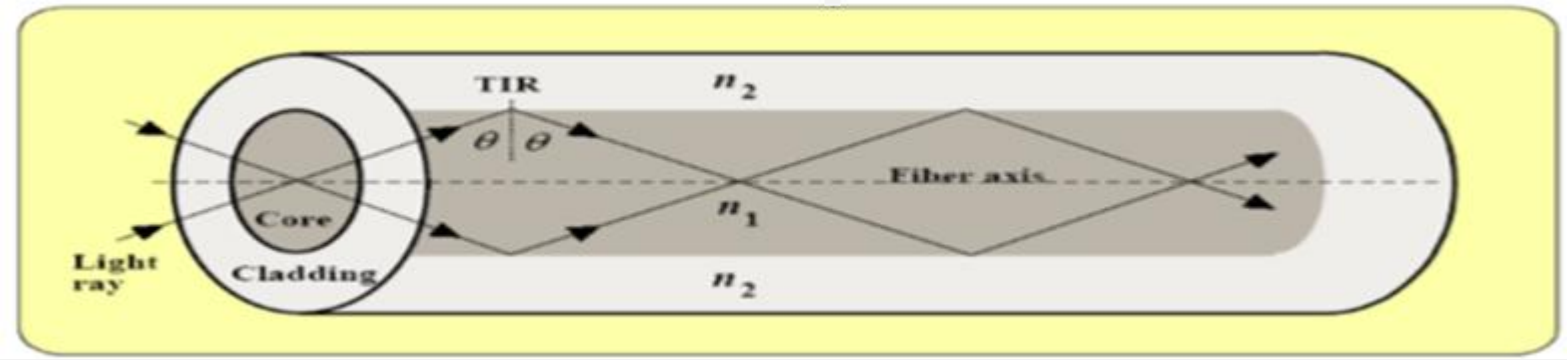
Light ray guiding condition

- To save light
  - inside a strand of fiber, we need to have it strike the
    - core-cladding boundary at an angle above the critical incident angle, θ_c, in order to provide total reflection of this light



Light ray guiding condition

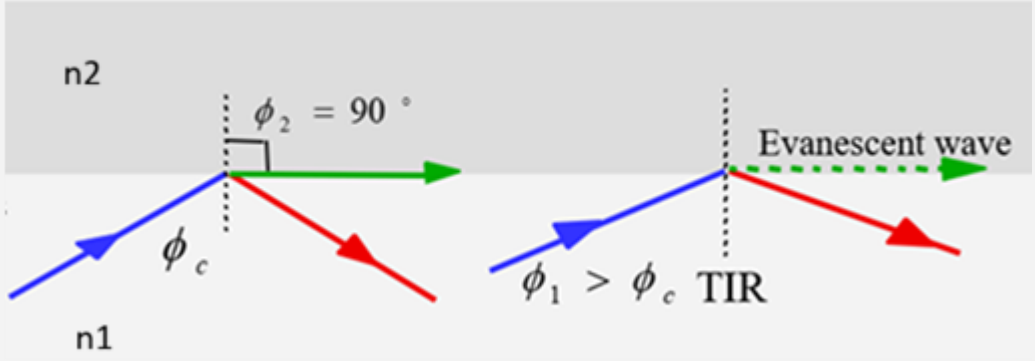
- Light ray that satisfies
  - total internal reflection condition at the Interface/boundary of the CORE and the CLADDING can be guided along an optical fiber



Light ray guiding condition

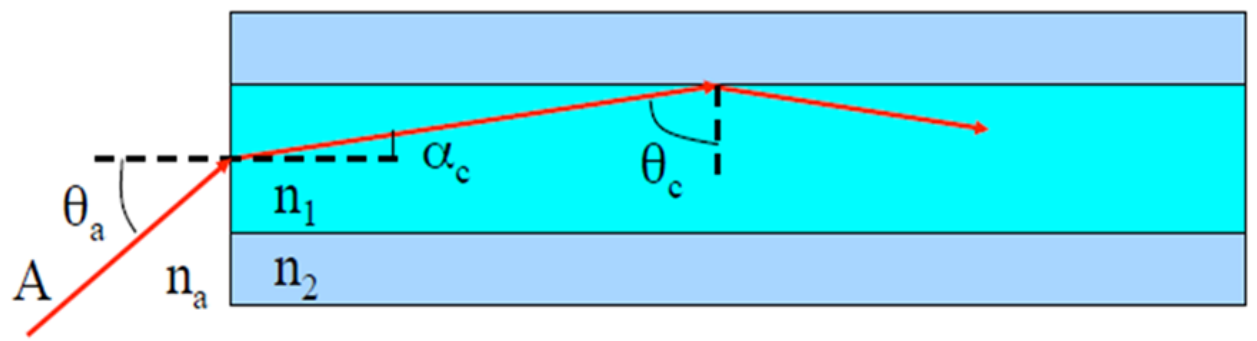
- Under what condition will light be trapped inside the fiber core?
  i. $n_1 = 1.48$ and $n_2 = 1.46$ (silica)
  ii. $n_1 = 1.495$ and $n_2 = 1.402$ (Plastic)

$$\phi_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$



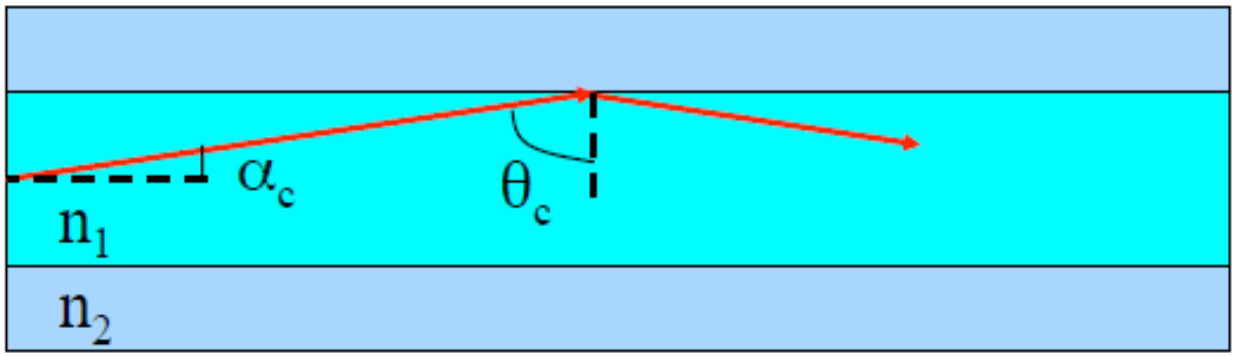
Light ray guiding condition

- Total Internal Reflection (TIR)
  - Critical incident angle
  - Critical propagation angle
- Launching of light
  - Acceptance angle
  - Numerical aperture



Critical- Incident & Propagation angles

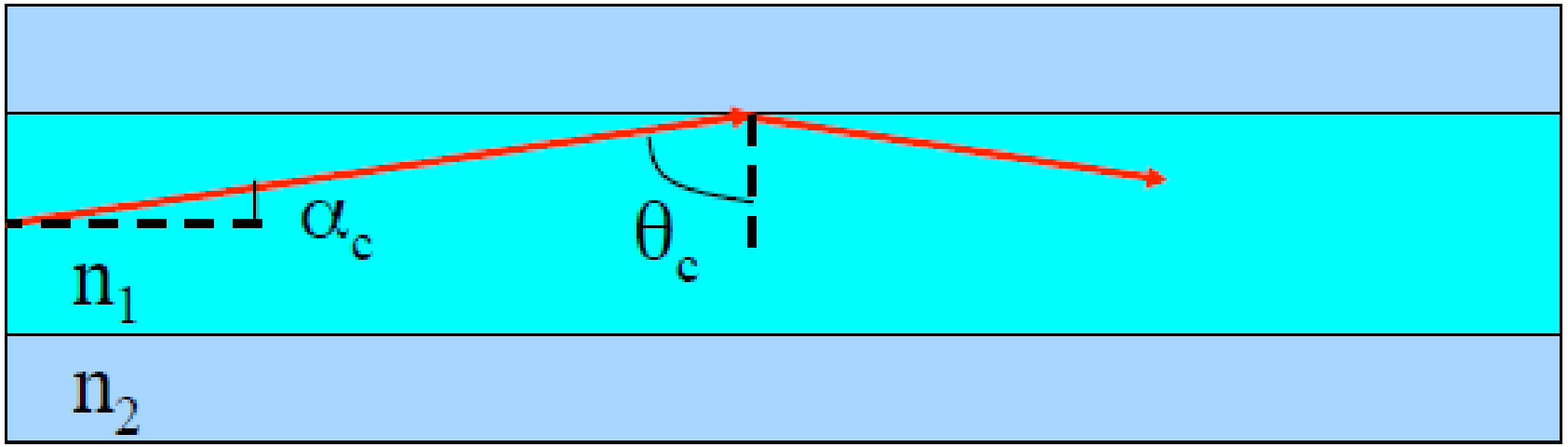
- Two key terms
  i. Critical incident angle, θ_c
     - Is the angle the beam makes with the line perpendicular to the boundary between the core and the cladding
  ii. Critical propagation angle, α_c
     - Is the angle the beam makes with the centerline of the optical fiber



Critical- Incident & Propagation angles

- Two key terms
  - Critical incident angle, θ_c
  - Critical propagation angle, α_c

$$\alpha_c = 90 - \theta_c$$



Critical- Incident & Propagation angles

- We know that

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

- Since

$$\alpha_c = 90 - \theta_c$$

- It implies

$$\cos\alpha_c = \sin\theta_c$$

Critical- Incident & Propagation angles

- thus

$$\sin^2 x + \cos^2 x = 1$$

$$\cos\alpha_c = \frac{n_2}{n_1}$$

$$\sin\alpha_c = \sqrt{1-\left(\frac{n_2}{n_1}\right)^2}$$

$$\alpha_c = \sin^{-1}\sqrt{\left(1-\left(\frac{n_2}{n_1}\right)^2\right)}$$

Critical- Incident & Propagation angles

- The critical propagation angle α_c
  - represents the requirement of achieving total internal reflection
- What is the critical propagation angle α_c
  - For $n_1 = 1.48$ and $n_2 = 1.46$
- State the requirement
  - of achieving total internal reflection in terms of critical propagation angle α_c

Critical- Incident & Propagation angles

- For n_1 = 1.48 and n_2 = 1.46

$$\alpha_c = \sin^{-1}\sqrt{\left(1-\left(\frac{1.46}{1.48}\right)^2\right)} = 9.43^\circ$$

- What happens
  - when a beam travels at a propagation angle of 10 degrees?

Critical- Incident & Propagation angles

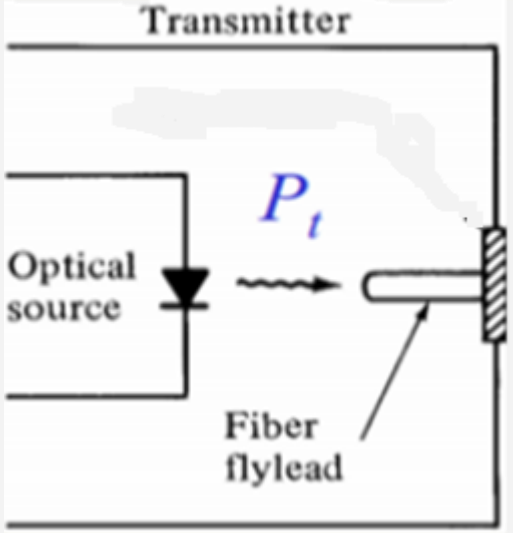
- In conclusion,
  - To save light
    - inside a strand of fiber, we need to have it strike the
      - core-cladding boundary at an angle above the critical incident angle, θ_c, in order to provide total reflection of this light
        - To make light fall above that angle, we have to direct it so that it is below the critical propagation angle, α_c ,with respect to the centerline of the fiber

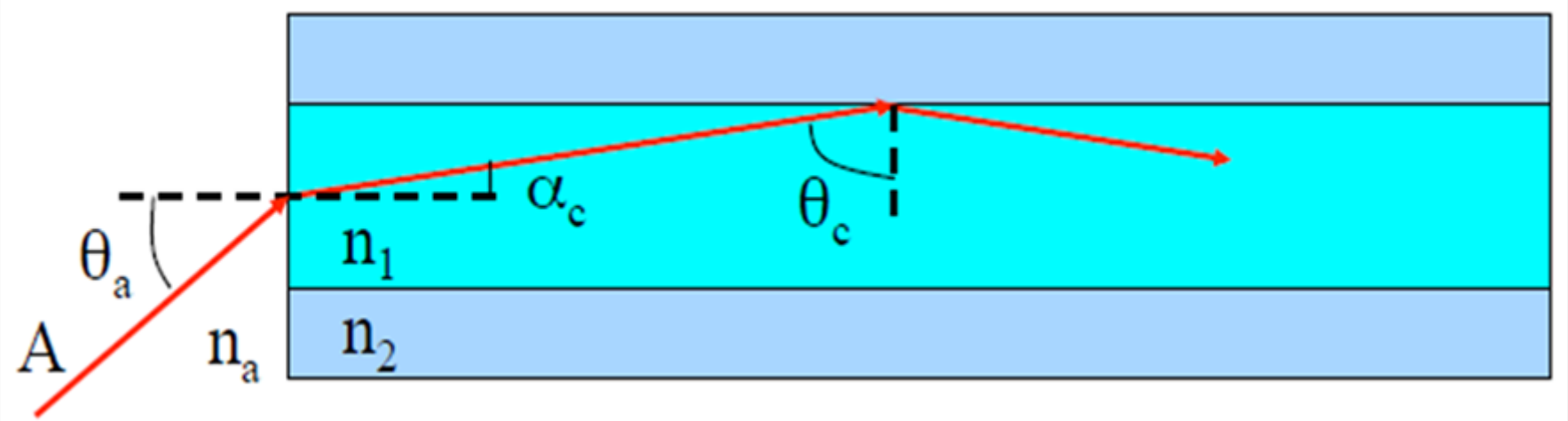
Acceptance Angle

- The next question that arises is
  - How can we direct this beam so that
    - it does indeed fall below the critical propagation angle?

Acceptance Angle

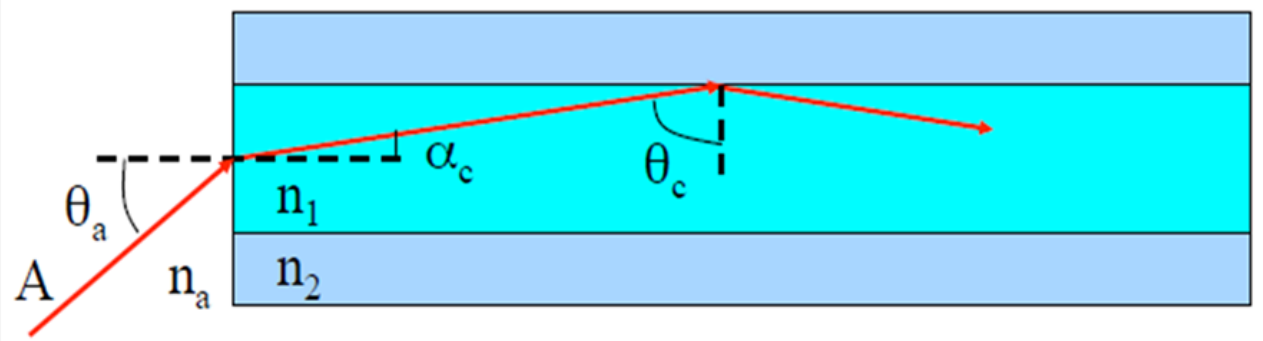
- The light
  - of course, must come from a source that is outside the fiber and therefore have to be directed into the fiber





Recap

- Light ray guiding condition
  - Total Internal Reflection (TIR)
    - Critical incident angle
    - Critical propagation angle
  - Launching of light
    - Acceptance angle
    - Numerical aperture

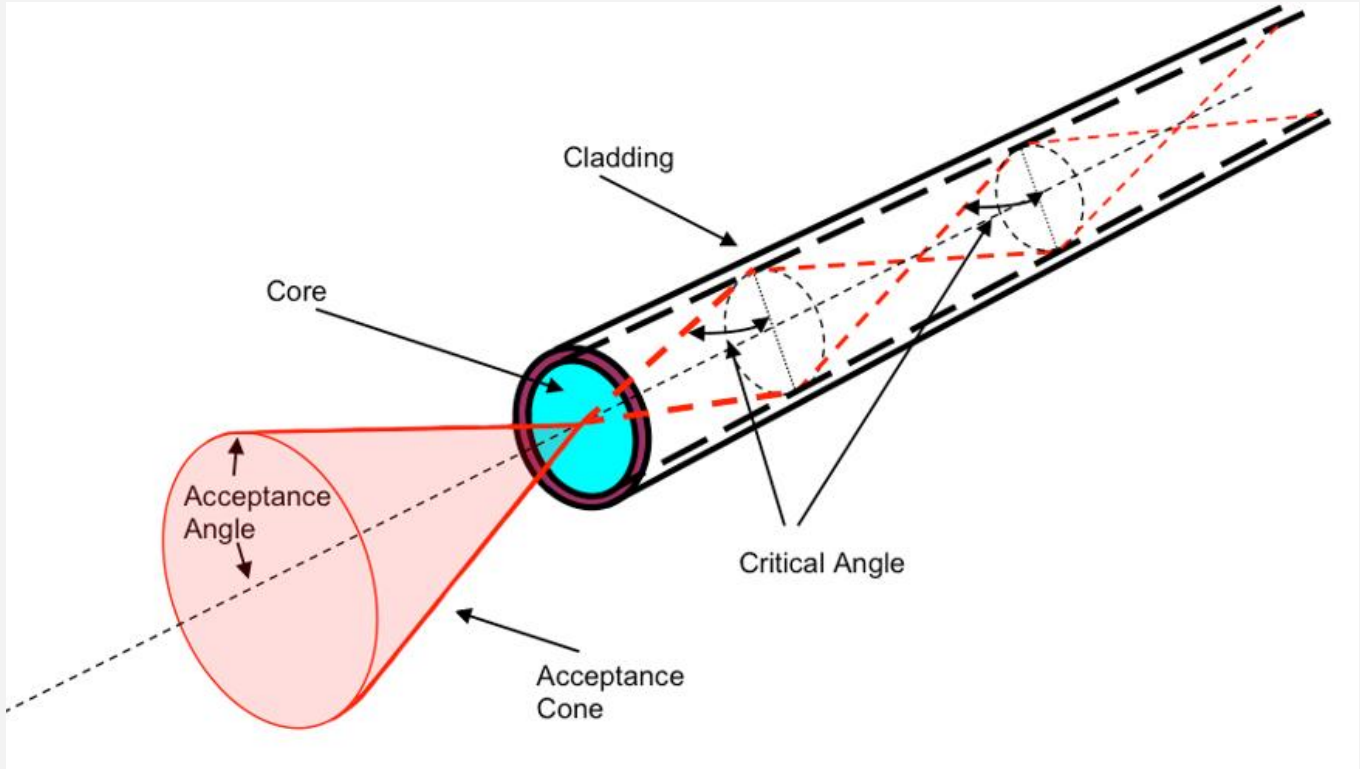


Acceptance Angle

- To save light
  - inside a fiber all rays must propagate at an angle less than the critical propagation angle α_c
    - In order for light to be maintained in the fiber at this angle, it must be directed from outside the fiber at angle θ_a or less

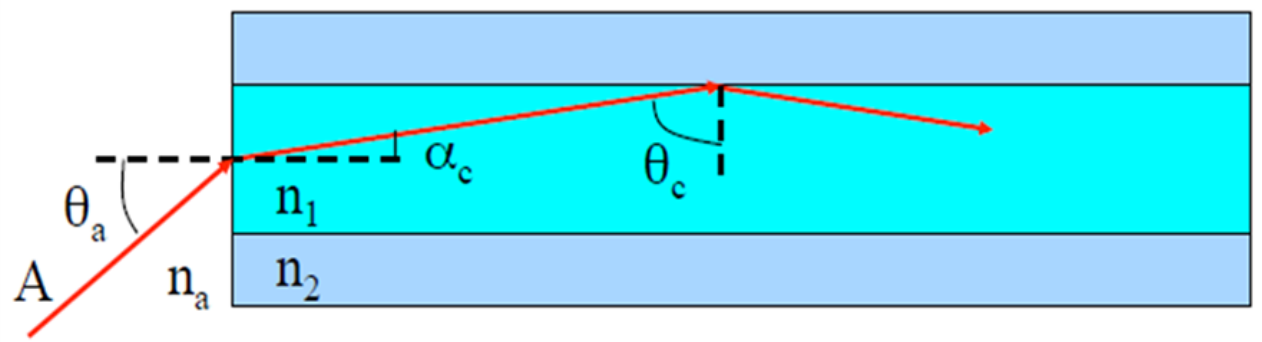
Acceptance Angle

- Light will be
  - saved inside the fiber if it comes from a source bounded by the cone $2\theta_a$ (cone of light)



Acceptance Angle

- Launching light into an optical fiber
  - at the gap-fiber interface, the beam at angle θ_a is the incident beam and the beam at angle α_c is the launched one

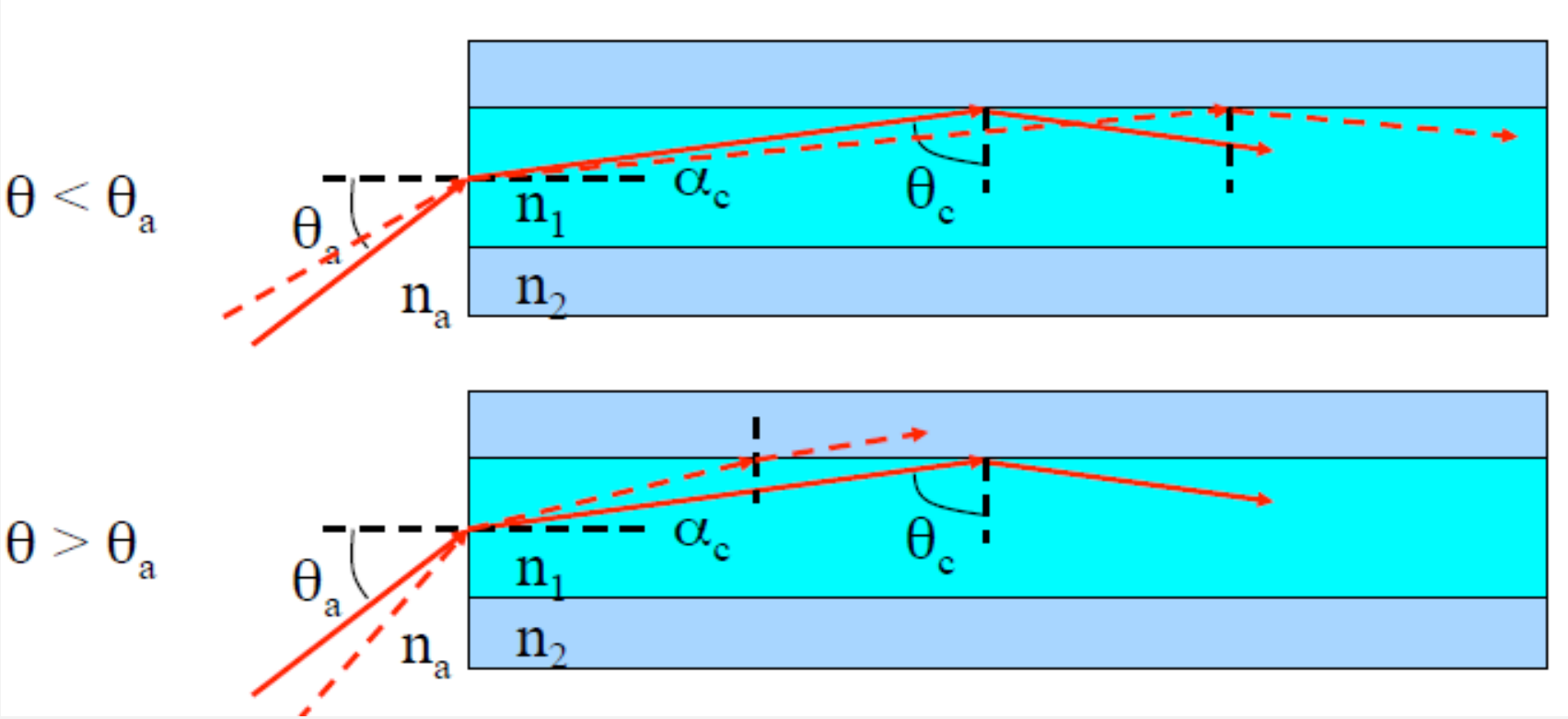


- The relationship between
  - θ_a and α_c can be derived using Snell's law

$$n_a \sin\theta_a = n_1 \sin\alpha_c$$

Acceptance Angle

- What happens when the incidence beam, θ is less than θ_a ?
- What happens when the incidence beam, θ is greater than θ_a ?

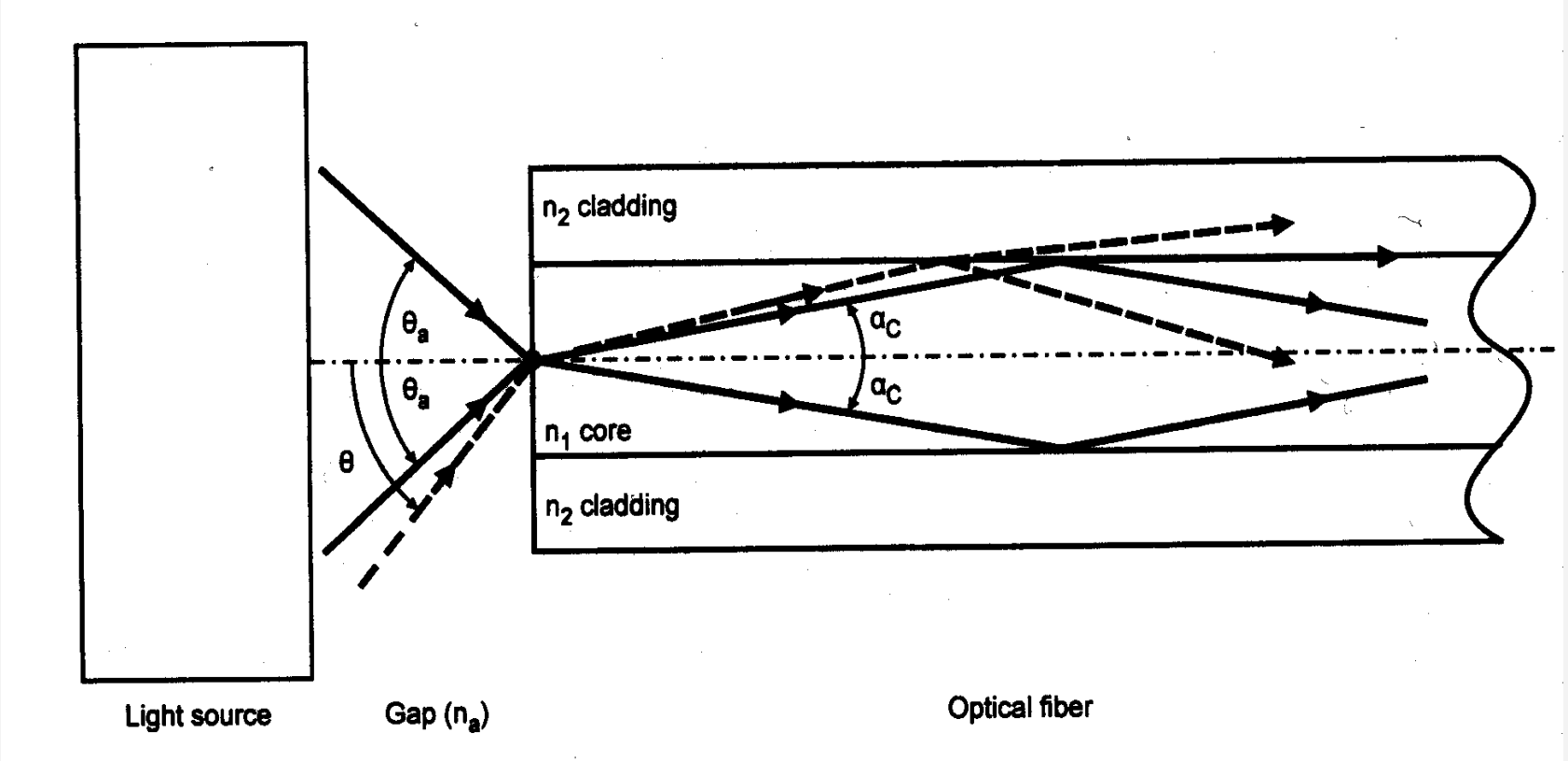


Acceptance Angle

- Any ray which is incident into the fiber core at
  - an angle greater than θ_a will have a propagation angle greater than α_c and therefore incident angle less than θ_c at the core-cladding interface and will NOT be totally internal reflected

Acceptance Angle

- Thus θ_a defines
  - an acceptance cone for an optical fiber



Acceptance Angle

- Light will be
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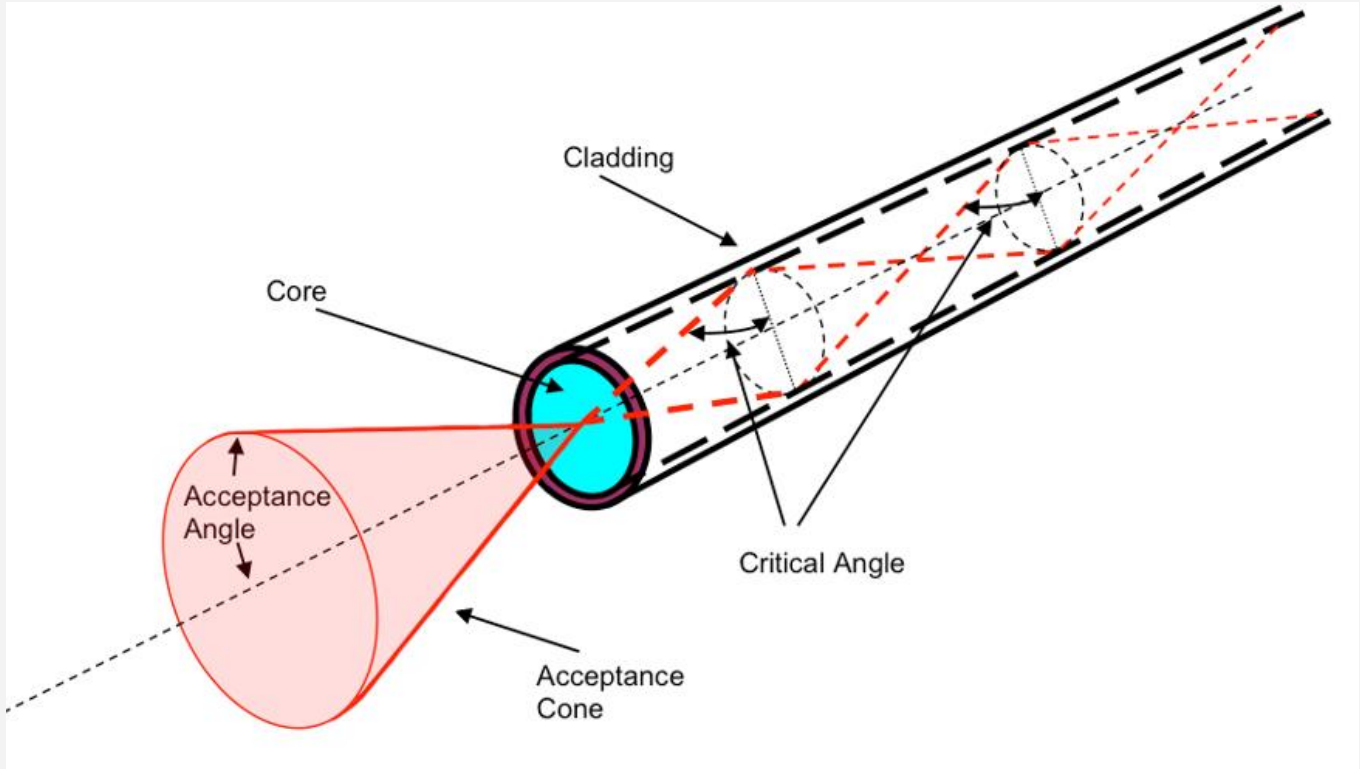


Acceptance Angle

- To save light
  - inside a fiber all rays must propagate at an angle less than the critical propagation angle α_c
    - In order for light to be maintained in the fiber at this angle, it must be directed from outside the fiber at angle θ_a or less

Acceptance Angle

- What
  - is the acceptance angle and cone of light for a fiber when $n_1 = 1.48$ and $n_2 = 1.46$

Acceptance Angle

- What
  - is the acceptance angle and cone of light for a fiber when $n_1 = 1.48$ and $n_2 = 1.46$

$$\sin\theta_a = n_1 \sin\alpha_c$$

$$\sin\alpha_c = \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

Acceptance Angle

- The acceptance angle is
  - described using Numerical Aperture (parameter in data sheets)

Numerical Aperture (NA)

- NA describes
  - the ability of an optical fiber to gather light from a source and
    - the ability to preserve or save, this light inside the fiber because of total internal reflection

$$NA = \sin\theta_a$$

Numerical Aperture (NA)

- We know that

$$\sin\theta_a = n_1 \sin\alpha_c$$

- But

$$\sin\alpha_c = \sqrt{1-\left(\frac{n_2}{n_1}\right)^2}$$

- therefore

$$NA = n_1 \sin\alpha_c = n_1\sqrt{1-\left(\frac{n_2}{n_1}\right)^2} = \sqrt{(n_1)^2-(n_2)^2}$$

Numerical Aperture (NA)

- It is very common
  - to use the relative difference of the refractive indexes, Δ, often called the relative index
- Thus

$$\Delta = \frac{n_1 - n_2}{n_1}$$

- We assume

$$n_1 \approx \frac{n_1 + n_2}{2}$$

Numerical Aperture (NA)

- The numerical aperture

$$NA = \sqrt{(n_1)^2 - (n_2)^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)}$$

- But

$$(n_1 - n_2) = n_1 \Delta$$

$$(n_1 + n_2) = 2 n_1$$

- thus

$$NA \approx \sqrt{n_1 \Delta \cdot 2 n_1} \approx n_1 \sqrt{2\Delta}$$

Problem

- Calculate the numerical aperture for the following two fibers
  i. $n_1 = 1.48$ and $n_2 = 1.46$ (silica)
  ii. $n_1 = 1.495$ and $n_2 = 1.402$ (Plastic)

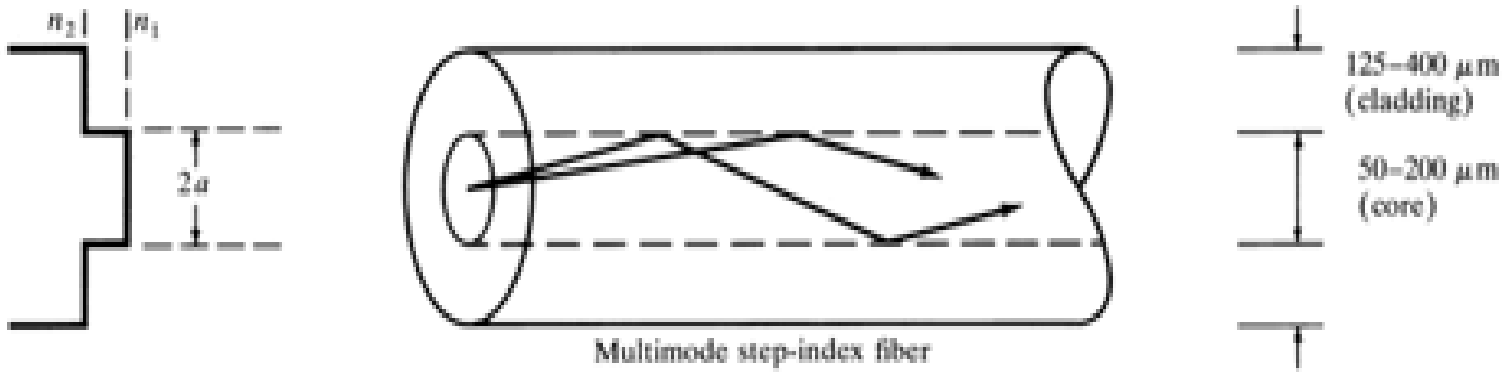
- Comment on your answers

Problem

- Briefly explain the phenomenon called total internal reflection and its importance in fiber optic communication technology
- Explain what is meant by the following terms and their importance in fiber-optic communication technology
  - Critical incident angle
  - Critical propagation angle
  - Acceptance angle
  - Numerical aperture
- For specific fiber, NA = 0.275 and n_1= 1.49
  - Find the critical propagation angle

Type of Fiber

- So far
  - The basic type of fiber optic we encounter is:
  1. Step-index multimode



Multimode step-index fiber

Focus

- Objectives
  1. To gain insight into the propagation of light along an optical fiber
  2. To explain some phenomena that light goes through
     a. Attenuation (Reduction in power level with distance)
        - Definition
        - Mechanisms
        - Loss calculation
     b. Dispersion (eroding of clarity with distance and speed)
        - definition
        - Concept of modes
        - Calculation of number of modes
        - Multimode fiber
          - Modal (intermodal) dispersion (cause)
            - Pulse spread calculation

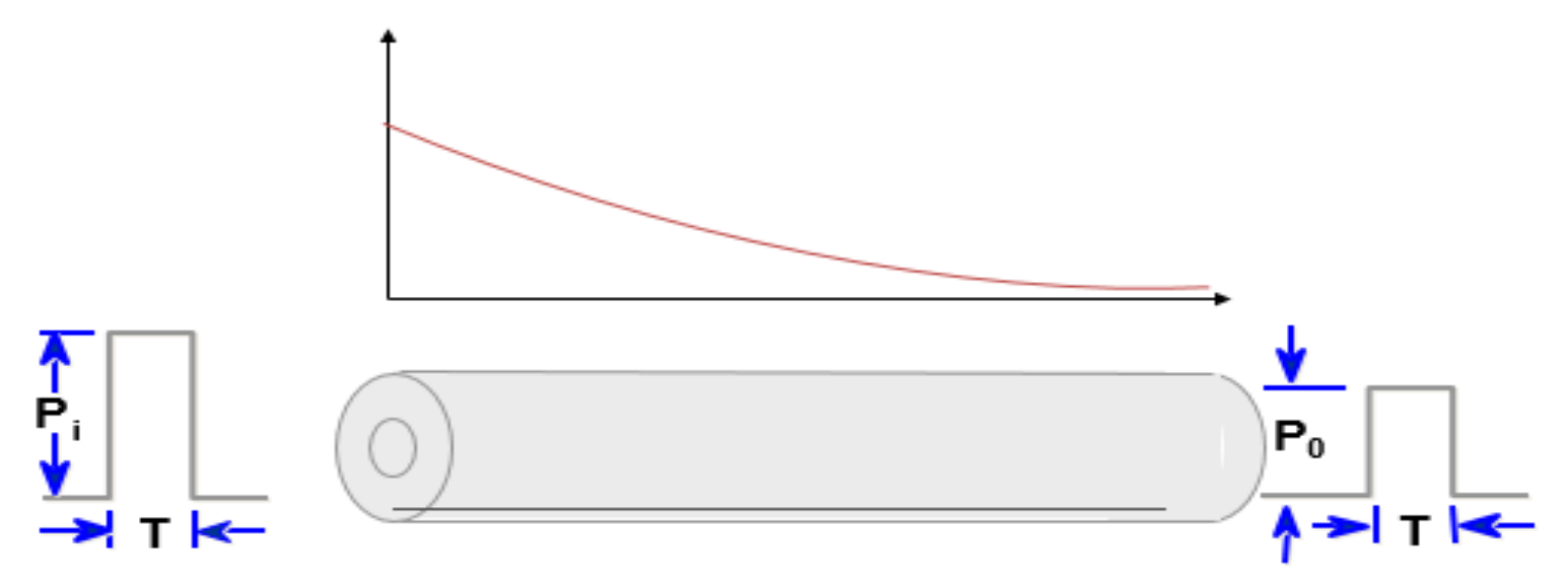
Attenuation and Dispersion

- This section
  - describes the operational characteristics of fibers
- Thus
  1. What are the loss or signal attenuation mechanisms in a fiber?

  2. Why and to what degree do optical signals get distorted as they propagate along a fiber?

Attenuation - Definition

- Signal attenuation
  - In fiber-optic communications
    - attenuation is the decrease in light power during light propagation along an optical fiber



Attenuation - Definition

- Signal attenuation
  - is one of the most important properties of an optical fiber
    - Because it largely determines the maximum unamplified or repeater-less separation between a transmitter and a receiver

Attenuation - Mechanisms

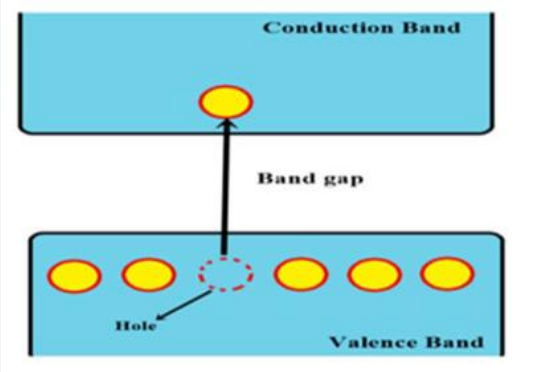
- The basic attenuation mechanisms in a fiber are
  - Intrinsic losses
    - i. Absorption
    - ii. Scattering
  - Extrinsic losses
    - Bending (micro-bending and macro-bending)

Mechanisms - Absorption

- Recall that
  - If an incoming photon has a frequency, f, wavelength λ, such that its energy,

$$E_P = hf = \frac{hc}{\lambda}$$

  - is equal to the energy gap of the material, the photon will be absorbed by the material



Mechanisms - Absorption

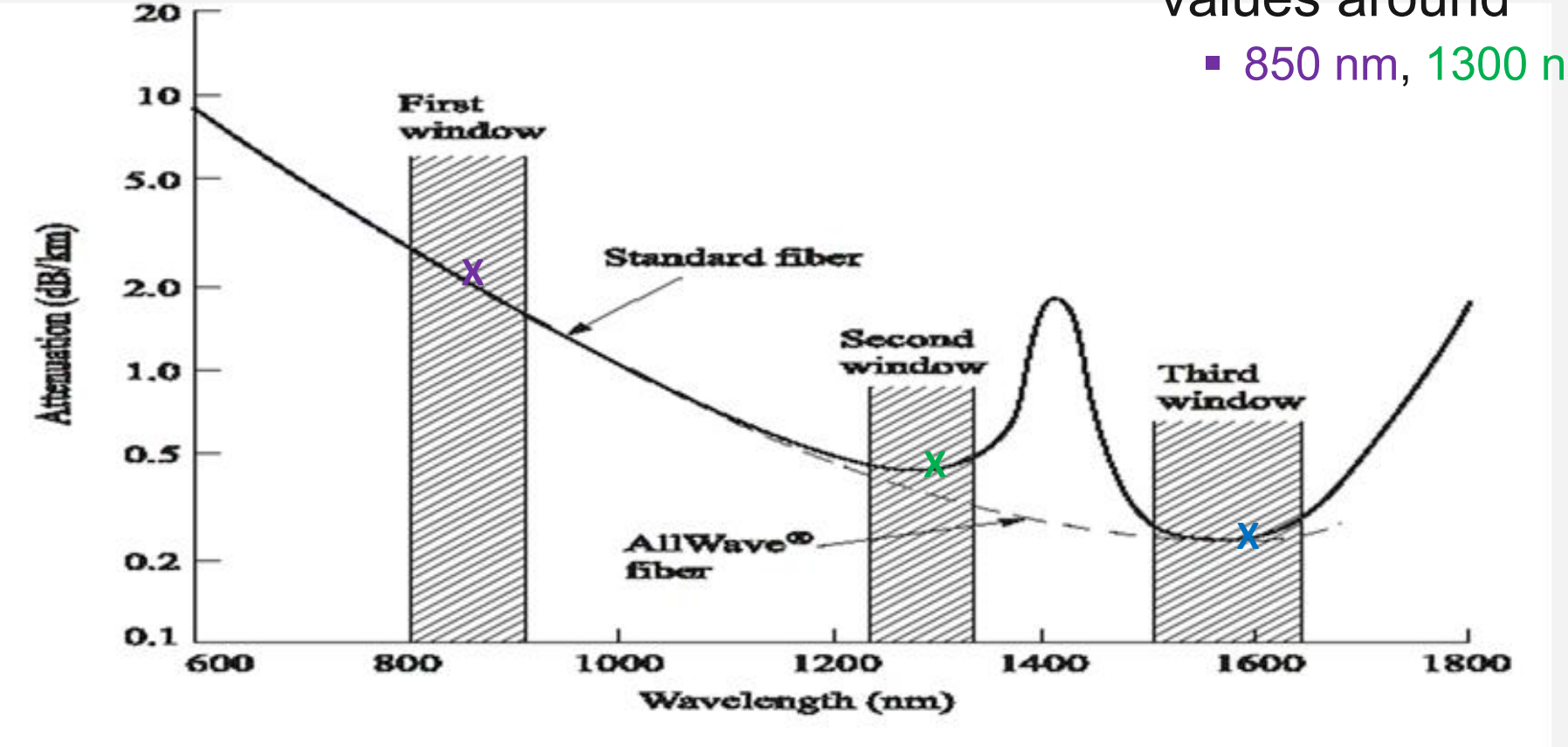
- When
  - light travels down an optical fiber and encounters a material whose energy gap is exactly equal to the energy of
    - these photons, the light will be absorbed
      - This type of loss is wavelength dependent

Mechanisms - Absorption

- Optical fiber
  - is a transparent strand, that is ‘non absorptive’ material
    - Absorption properties that still remain are caused not by silica atoms but by some molecules of the hydroxide anion OH^-
      - These molecules are incorporated in silica during the fabrication process and is very hard to eliminate them
- OH^- molecules have major peaks of absorption at
  - 945 nm, 1240 nm, and 1380 nm

Spectral – Attenuation Curve

- Attenuation reaches minimum values around
  - 850 nm, 1300 nm, and 1550 nm



Transmission Spectral (Transparent) Windows

- Three major fiber transmission spectral windows

The 1st window: 850 nm, attenuation 4 dB/km

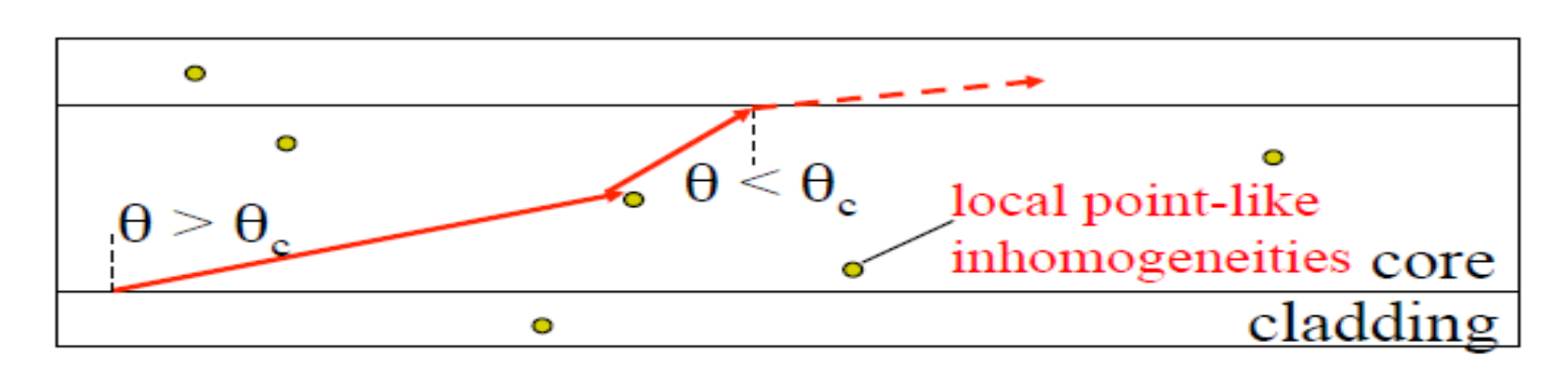
The 2nd window: 1300 nm, attenuation 0.5 dB/km

The 3rd window: 1550 nm, attenuation 0.2 dB/km

1550 nm window is today's standard **long-haul** communication wavelengths.

Mechanism - Scattering Loss

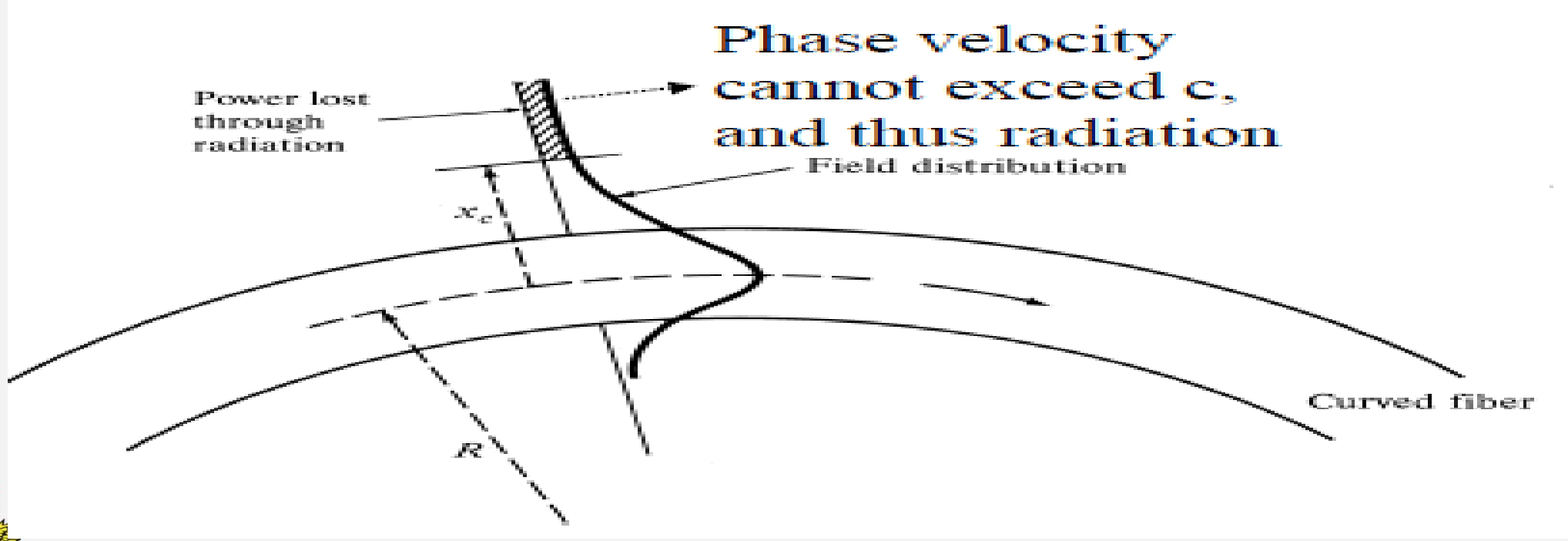
- When there is an imperfection in the core material
  - a beam propagating at the critical propagation angle or less changes direction after it meets an obstacle
    - That is light will be scattered



    - Imperfection means
      - slight variations in the refractive index

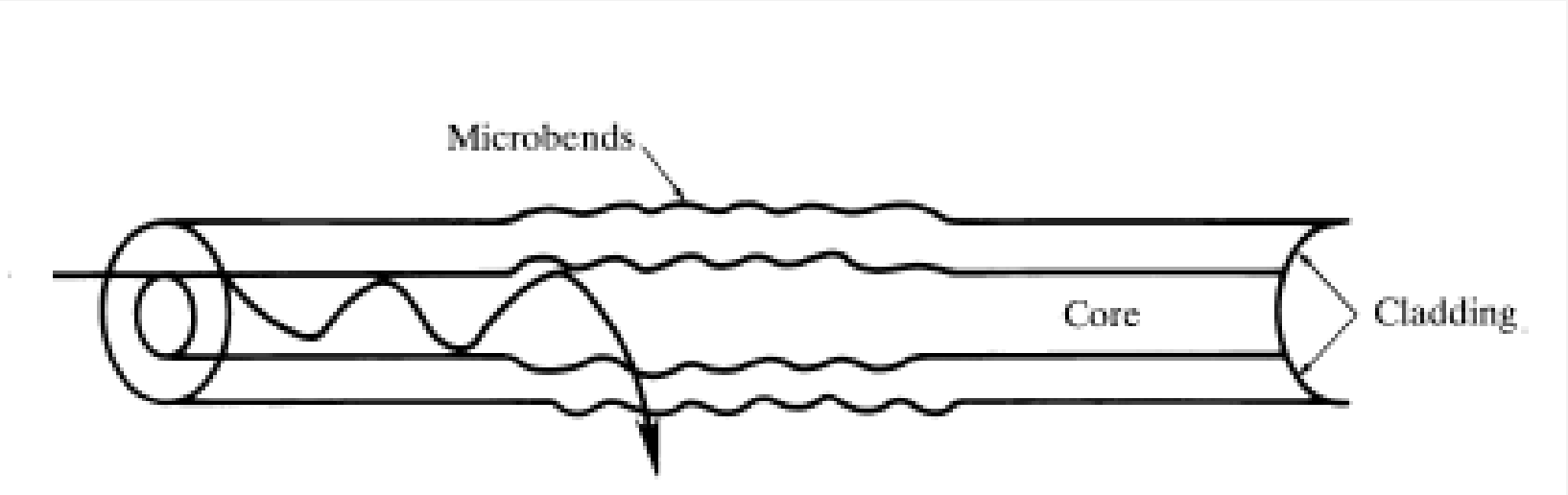
Mechanism - Bending loss

- Macro-bending
  - Loss caused by the curvature of the entire axis



Mechanism - Bending loss

- Micro-bending
  - Loss caused by micro-deformations of the fiber axis



- Bending of the fiber
  - leads to disruption of the condition of total internal reflection

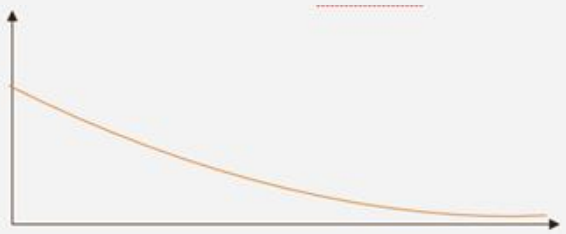
Optical Attenuation

- Signal amplitude reduction limits “how far”
- Attenuation in dB
- Power is measured in dBm:

Calculation of total Attenuation

- Fiber loss is
  - the ratio of power at the output end of fiber, P_{out}, to power launched into the fiber, P_{in}

$$Loss = \frac{P_{out}}{P_{in}}$$



  - where power is measured in watts

Calculation of total Attenuation

- In communication technology
  - Loss (attenuation) is measured in decibels (dB)

$$Loss(dB) = -10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

  - The negative is introduced
    - in order to present loss as a positive quantity

Calculation of total Attenuation

- Attenuation per unit of fiber length, A is

$$A(dB/km) = Loss(dB)/fiberlength(km)$$

$$A(dB/km) = -\frac{10}{L}\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

- The quantity, *A (dB/km)*
  - is called attenuation and it is one of the most important characteristics of an optical fiber

Calculation of total Attenuation

- From the attenuation equation we have

$$P_{out} = P_{in} 10^{\frac{-AL}{10}}$$

- Pre-repeater distance

$$L = \frac{10}{A} \log_{10}\left(\frac{P_{in}}{p_{out}}\right)$$

Problem

- Calculate the maximum transmission distance for a fiber link with an attenuation of 0.5 dB/km if the power launched is 1 mW and receiver sensitivity is 50 μW

Problem

- A certain optical fiber has an attenuation of 0.6 dB/km at 1310 nm and 0.3 dB/km at 1550 nm. Suppose the following two optical signals are launched simultaneously into the fiber:
  a. An optical power of 150 μW at 1310 nm
  b. An optical power of 100 μW at 1550 nm
- What are the power levels in μW of these two signals at
  i. 8 km
  ii. 20 km
- Comment on your answer

Attenuation and Dispersion

- This section
  - describes the operational characteristics of fibers
- Thus
  1. What are the loss or signal attenuation mechanisms in a fiber?

  2. Why and to what degree do optical signals get distorted as they propagate along a fiber?

Recap-Attenuation

i. What is attenuation in an optical fiber?
ii. What is loss in optical fiber?
iii. What does the term transparent window mean?

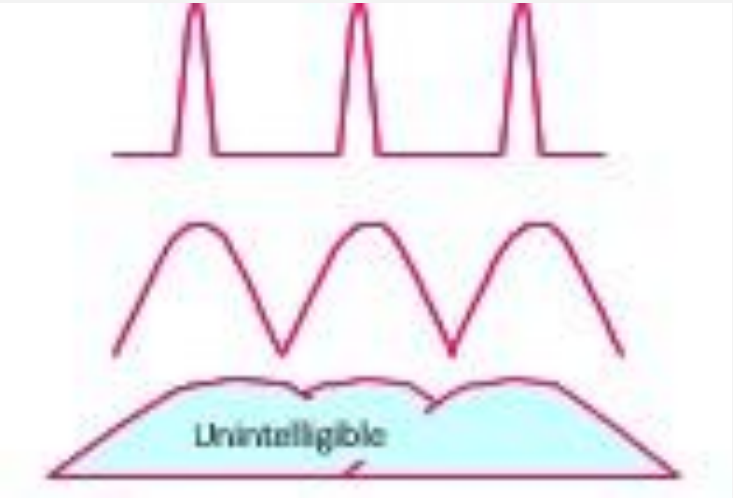
Dispersion in Optical Fiber

- Why and to What
  - degree do optical signals get DISTORTED as they propagate along a fiber
- Fiber dispersion
  - results in optical pulse broadening and hence digital signal degradation

Dispersion in Optical Fiber

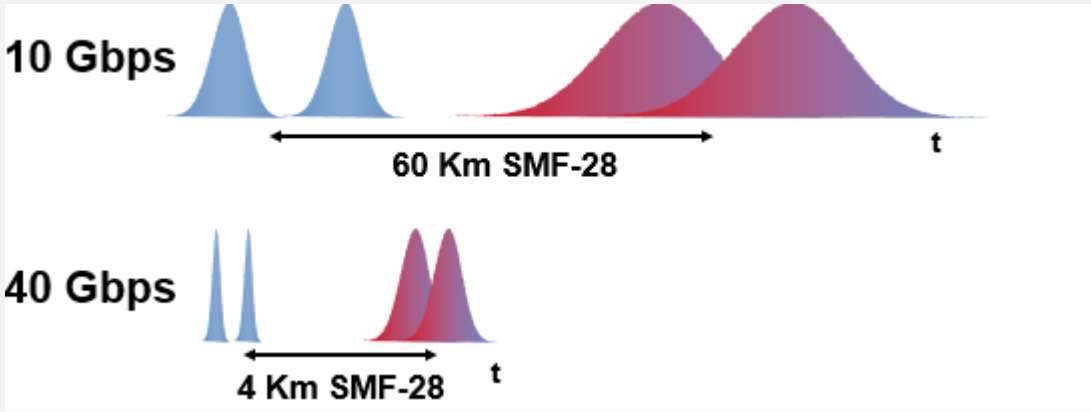
- Successive pulses overlap as they spread
- Spreading increases with distance
- Degree of dispersion depends on fiber type

- Initial instantaneous pulses



Dispersion- Limitations

- Dispersion causes pulse distortion, pulse "smearing" effects
- Limits "how fast“ and “how far”



Dispersion in Optical fiber

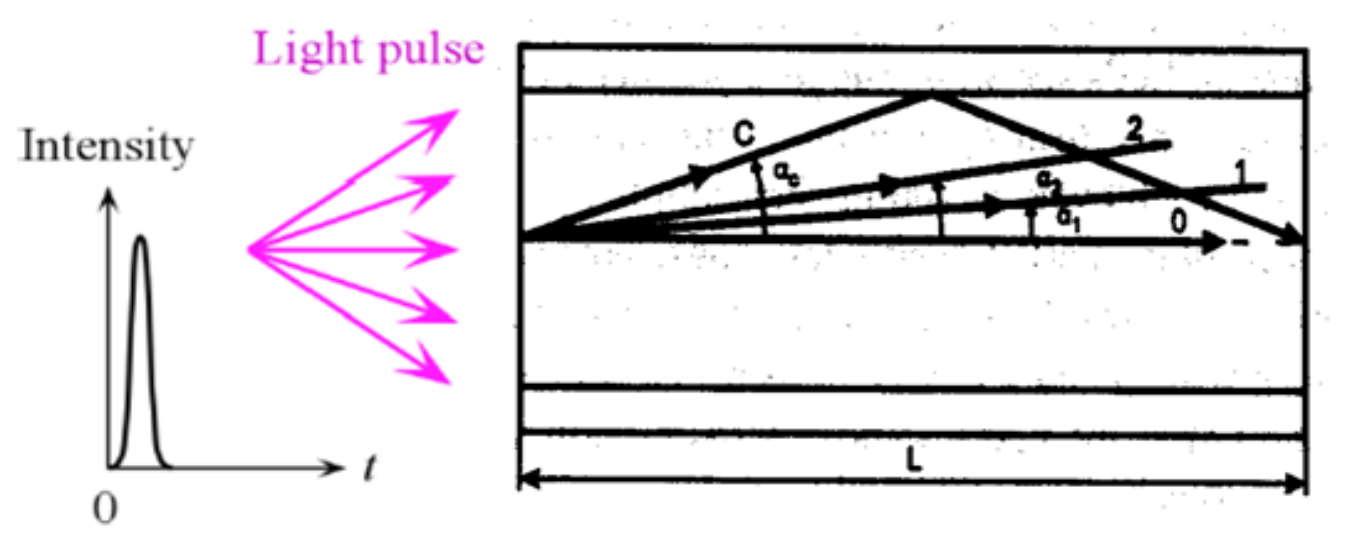
- Types of fiber dispersion
  1. Modal (intermodal) dispersion
  2. Chromatic dispersion
  3. Polarization Mode dispersion

Mode Concept

- This section
  - presents an overview of the underlying concepts of optical fiber modes

Mode Concept

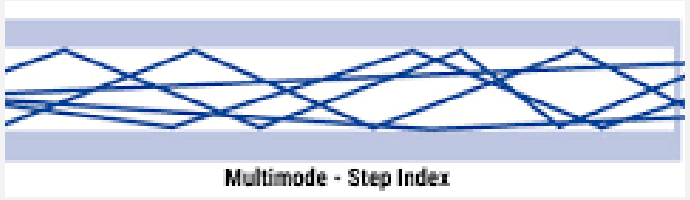
- Light entering an optical fiber
  - ‘breaks’ down into discrete beams called MODES
    - Stable electromagnetic field patterns existing within the fiber



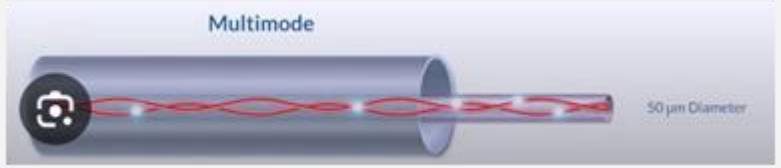
- Optical fiber modes are the
  - specific, distinct paths or electromagnetic field patterns that light rays take as they propagate through an optical fiber

The power of launched light is delivered by separate modes within the fiber

Mode Concept

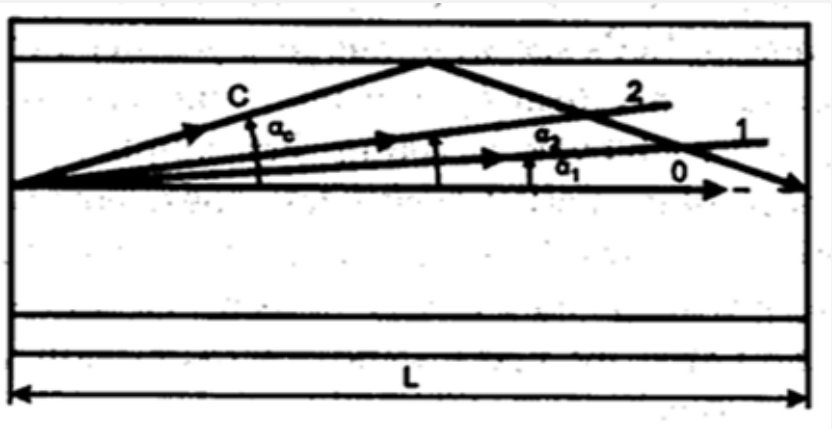


- Each mode or pattern
  - can be seen as a separate beam traveling at a certain propagation angle ranging from zero to α_c
- The mode traveling
  - precisely along the fiber central axis is the zero-order mode or fundamental mode
  - at the critical angle is the highest mode
- Many modes
  - can exist within a fiber, and so a fiber having many modes is called a multimode fiber



Mode Concept

- The propagation of
  - light along a waveguide can be described in terms of a set of guided electromagnetic waves



- Thus, the electromagnetic light field
  - that is guided along an optical fiber can be represented by a superposition of trapped modes

Mode Concept

- The power of launched light
  - is delivered by separate modes within the fiber
- At the end of the fiber
  - the modes combine to constitute the output pulse

Mode Concept

- Number of modes
  - How many modes an optical fiber can carry depends on the
    - optical and geometric characteristics of a fiber
      - core diameter, wavelength, and numerical aperture
- An important parameter
  - connected with the number of modes is the
    - normalized frequency parameter or V-parameter
- It serves as a fundamental relationship
  - between the physical parameters of the fiber (core radius, refractive index) and its operational performance (modes, cut-off).

Mode Concept

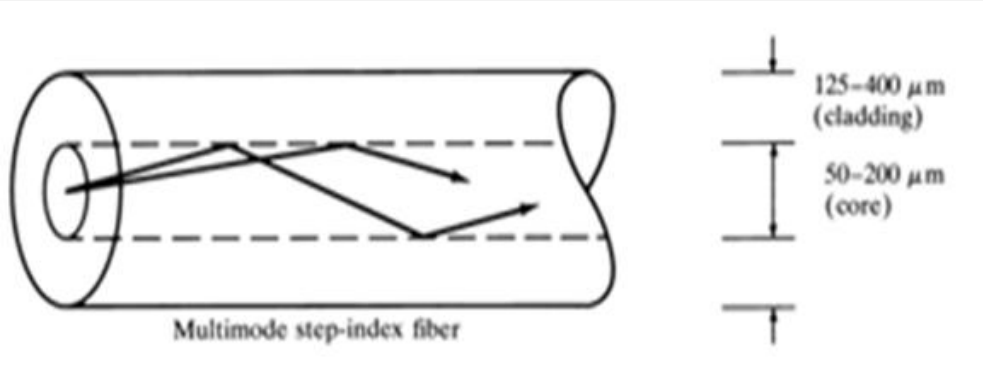
- The V-number is given by

$$V = \frac{\pi d}{\lambda}\sqrt{n_1^2 - n_2^2} = \frac{\pi d}{\lambda} NA = \frac{\pi d}{\lambda} n_1 \sqrt{2\Delta}$$

- where
  - d = the core diameter
  - λ = operating wavelength

V-number defines the

- mode propagation capacity of an optical fiber
- cutoff frequency for single-mode fibers and indicates the number of guided modes in multimode fibers



Mode Concept

- For a large V number (V > 20)
  - The number of modes for step-index fiber is

$$N = \frac{V^2}{2}$$

Example

- Suppose we have a multimode step-index optical fiber that has a core radius of 25 μm, a core index of 1.48 and relative refractive index of 1%.
  - What are the number of modes in the fiber at wavelength: 860, 1310 and 1550 nm

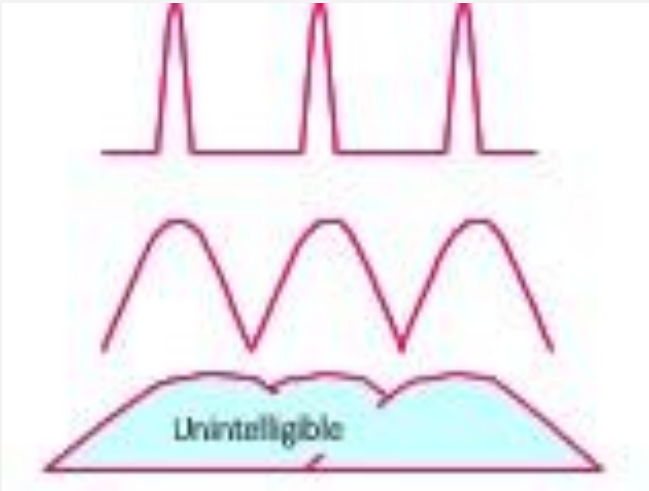
Problem

- A step-index multimode fiber with a numerical aperture of 0.20 supports approximately 1500 modes at 850-nm wavelength
  - What is the diameter of its core?
  - How many modes does the fiber support at
    - 1320 nm
    - 1550 nm
    - What is the essence of these mode numbers?

Dispersion in Optical Fiber

- Types of fiber dispersion
  1. Modal (intermodal) dispersion
  2. Chromatic dispersion
  3. Polarization Mode dispersion

Initial instantaneous pulses



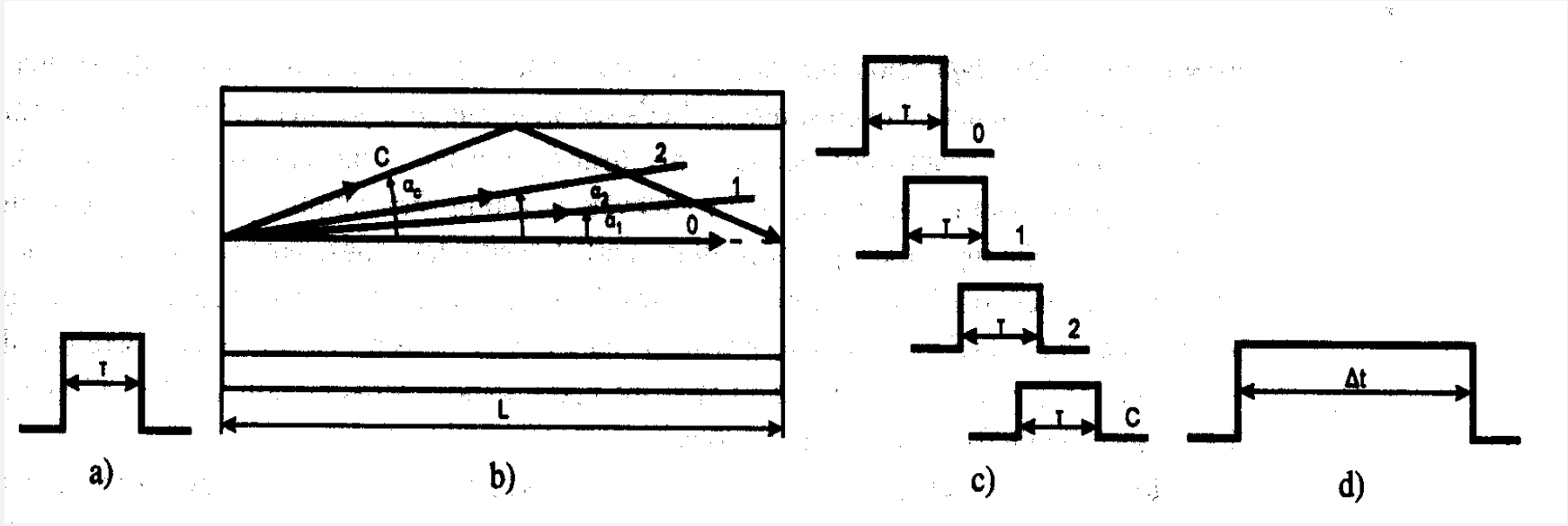
Modal (Intermodal) Dispersion

- The mode concept is of interest because
  - It explains modal (intermodal) dispersion

- Let's consider
  - A beam propagating inside a fiber taking into account the mode concept

Modal (Intermodal) Dispersion

- Light is used to carry communication signals
  - Such light pulse, radiated by a light source, enters fiber, where each pulse breaks into a set of small pulses carried by an individual mode



Modal (Intermodal) Dispersion

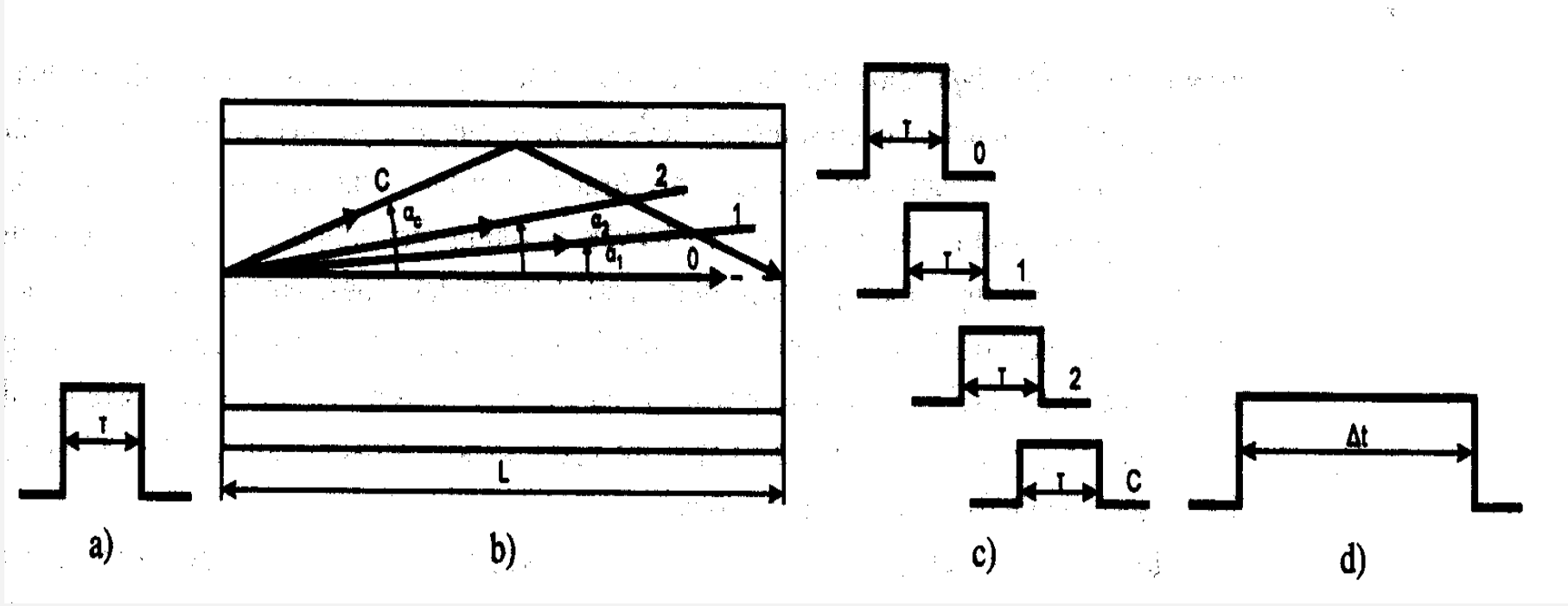
- At the fiber output
  - individual pulses recombine and since they arrive at different times and they overlap, the receiver sees one long light pulse
    - whose rising edge is from the fundamental mode and whose falling edge is from the critical mode

Modal (Intermodal) Dispersion

- Pulse broadening
  - caused by the mode structure of a light beam inside the fiber is called modal (intermodal) dispersion

- The pulse spread is
  - the difference in the arrival time of the pulse that
    - moves at the critical propagation angle and that which moves along the central axis

Pulse Spread Calculations



Pulse Spread Calculations

- Time needed by
  - a zero-order mode travelling along the central axis to reach the receiver end is

$$t_0 = \frac{L}{\upsilon}$$

  - where

    L = the link length

    $v = c/n_1$ is the light velocity within the core having refractive index n_1

Pulse Spread Calculations

- Time needed by
  - the highest-order mode propagating at the critical angle to reach the receiver end is

$$t_c = \frac{L}{\upsilon \cos \alpha_c}$$

  - But

$$\cos \alpha_c = \frac{n_2}{n_1}$$

Pulse Spread Calculations

- Thus

$$t_c = \frac{n_1 L}{n_2 \upsilon}$$

- Pulse broadening due to Intermodal dispersion is

$$\Delta t_{SI} = t_c - t_0$$

$$= \frac{n_1 L}{n_2 \upsilon} - \frac{L}{\upsilon}$$

Pulse Spread Calculations

- which becomes

$$\Delta t_{SI} = \frac{L}{\upsilon}\left[\frac{n_1}{n_2} - 1\right]$$

$$= \frac{n_1 L}{c}\left[\frac{n_1}{n_2} - 1\right]$$

$$= \frac{n_1 L}{c}\left[\frac{n_1 - n_2}{n_2}\right]$$

$$= \frac{n_1 L}{n_2 c}\left[n_1 - n_2\right]$$

Pulse Spread Calculations

- But

$$n_1 - n_2 = n_1 \Delta$$

- Thus

$$\Delta t_{SI} = \frac{n_1 L}{n_2 c}(n_1 \Delta) = \frac{L}{n_2 c} n_1^2 \Delta$$

- The pulse spread in nanoseconds is

$$\Delta t_{SI} = \frac{L}{2 n_2 c}(NA)^2$$

- Why silica not plastic?

Pulse Spread Calculations

- Note L is in km and c is in km/s

- Bit rate

$$bit\ rate = \frac{1}{\Delta t_{SI}} = \frac{2n_2 c}{L * NA^2}$$

Problem

- How much will a light pulse spread after traveling along 5 km of step-index fiber whose NA = 0.275 and n_1 = 1.48 and the relative refractive index is 1%

Problem

a. Show that in a step-index fiber the modal pulse spread per unit length can be approximated by

$$\Delta t_{SI} \Big/ L(km) = \frac{NA^2}{2n_2 C}$$

- where NA is the numerical aperture, n_2 the refractive index of the cladding and C, the velocity of light in vacuum

Problem

Consider a step-index fiber with core and cladding diameters of 62.5 and 125 μm, respectively. Let the core index n_1 = 1.48 and let the relative refractive index, Δ= 1.5%. Compare the modal dispersion in units of ns/km at 1310 nm of this fiber as given in (a) with the more exact expression

$$\Delta t_{SI} \Big/ L(km) = \frac{n_1 - n_2}{C}\left(1 - \frac{\pi}{V}\right)$$

Dispersion in Optical Fiber

- Types of fiber dispersion
  1. Modal (intermodal) dispersion
  2. Chromatic dispersion
  3. Polarization Mode dispersion

i. Physical reason(s)

ii. Pulse spread calculation

Problem

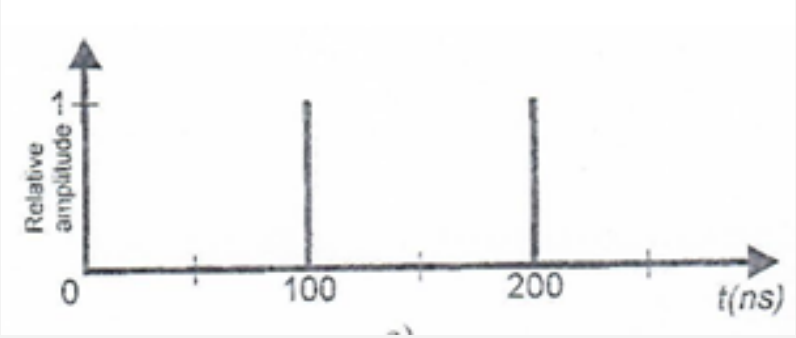
- How much will a light pulse spread after traveling along 5 km of step-index fiber whose NA = 0.275 and n_1 = 1.48 and the relative refractive index is 1%

How Intermodal Dispersion Restricts Bit rate

- Intermodal dispersion
  - restricts bit rate
- Suppose you
  - need to transmit information at 10 Mbps
    - What does this mean?

How Intermodal Dispersion Restricts Bit rate

- Suppose you
  - need to transmit information at 10 Mbps
    - This means you want to send 10×10^6 pulses every second
- Thus, a pulse every $1/(10 \times 10^6) = 100ns$
- Let's assume
  - that the duration of the input pulse is negligibly short



How Intermodal Dispersion Restricts Bit rate

- We are able
  - to distinguish pulses until they overlap
    - Thus, light carries information until the pulses overlap
- Problem
  - Find the maximum bit rate for the fiber discussed if the transmission length is 1 km

Bit rate limitation

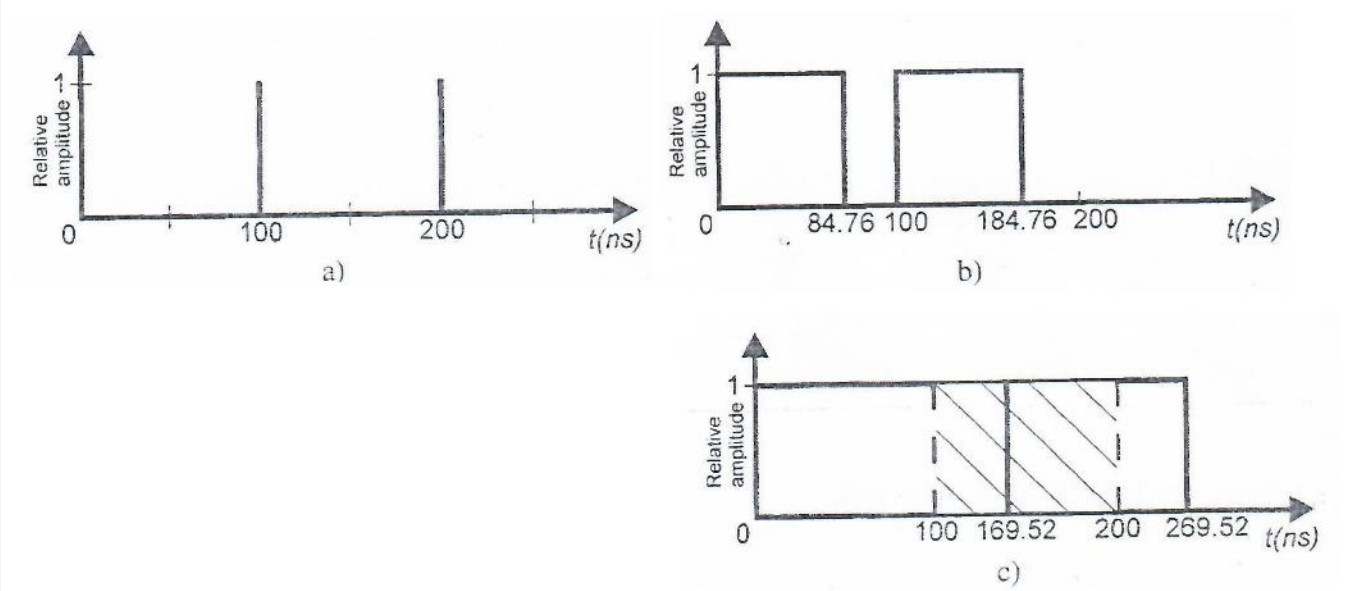
- Let Δt be
  - the width of an individual pulse
- Then a 1-second interval
  - can accommodate a certain number of these pulses before the overlap
    - This number is

$$number\ of\ pulses = \frac{1}{\Delta t}$$

Bit rate limitation

- In our example
  - the maximum bit rate is

$$\frac{1}{\Delta t} = \frac{1}{85.6\,ns} = 11.68\,Mbps$$



Bit rate limitation

- But
  - we want to have a time gap between adjacent pulses to ensure their separation
- Design consideration
  - even if pulses spread additional x %, they should not overlap

Bit rate limitation

- If we
  - take 25% of the cycle gap (even if pulses spread additional 25%, they should not overlap)
    - The maximum pulse width is

$$\Delta t + 0.25\Delta t = \Delta t \; x \, 1.25 = 85.6 \; x \, 1.25 = 107 \, ns$$

    - The maximum bit rate is

$$\frac{1}{107 \, ns} = 9.34 \, Mbps$$

    - Thus, we come up with a lower number for the maximum bit rate

Solutions to Modal Dispersion

- What is the physical reason for the problem?

Solutions to Modal Dispersion

- Physical reason for the problem
  - Within the core the zero-order mode travels along the central axis and the higher modes travel at, or less than the critical propagation angle
    - These beams travel at the same velocity but over different paths and arrive at the receiver end at different times and they overlap
- If we
  - could arrange it so that they will arrive simultaneously, we would solve the problem

First Solution

- Recall that
  - the velocity of light v, within a material is defined by its refractive index n as

$$\upsilon = \frac{c}{n}$$

- How can we use the above equation to solve the pulse spreading problem in multimode fibers?

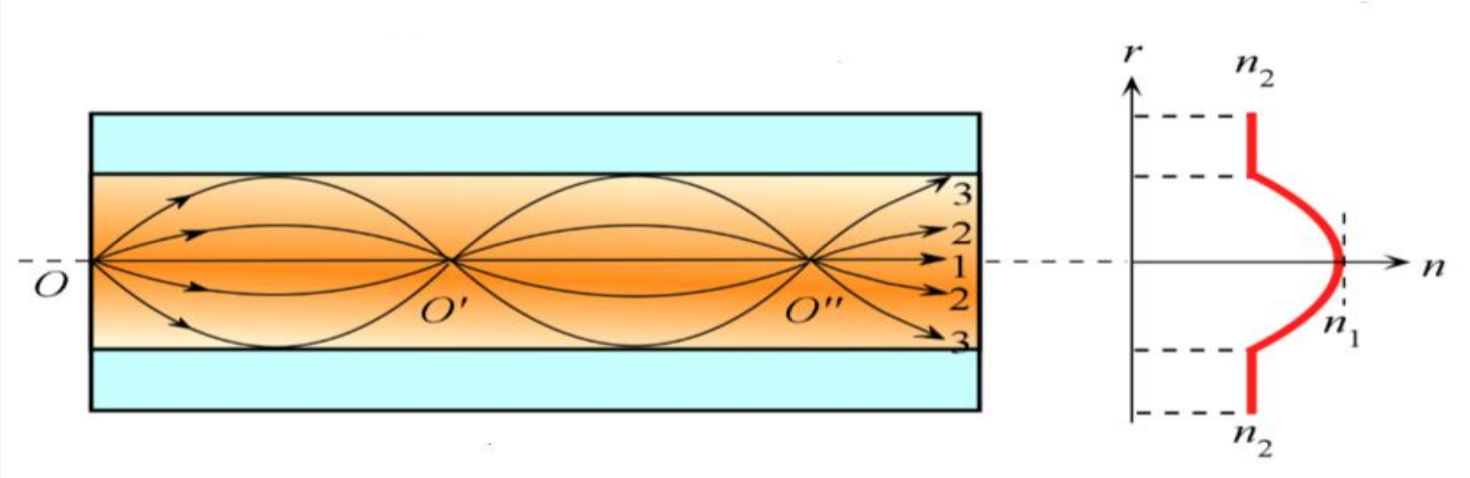
First Solution

- The solution approach
  - is to design the core with different refractive indexes so that
    - the beam traveling the farthest distance does so at the highest velocity and
      - the beam traveling the shortest distance propagates at the slowest velocity

- Such fibers are called
  - Graded-Index (GI) fibers

Graded-Index Fiber

- GI fiber
  - has a core material whose refractive index decreases continuously with distance from the fiber axis

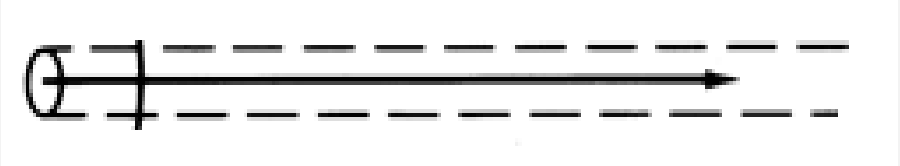


  - Ray paths
    - are different but so are the velocities along the paths so that the rays arrive at the same time

Graded-Index Fiber

- The numerical aperture is given by

$$NA = n_1 (2\Delta)^{1/2} \sqrt{1 - \left(\frac{r}{a}\right)^{\alpha}}$$



- r is the radial distance from the fiber axis
- a is the core radius
- α defines the shape of the index profile

Graded-Index Fiber

- r is the radial distance from the fiber axis
- a is the core radius
- n_1 is refractive index at the core axis
- n_2 is the refractive index of the cladding
- α defines the shape of the index profile

Graded-Index Fiber

- The pulse spread in GI fiber is given by

$$\Delta t_{GI} = \frac{n_1 L \Delta^2}{8c}$$

  - n_1 is the core group index of refraction
- the number of modes is given by

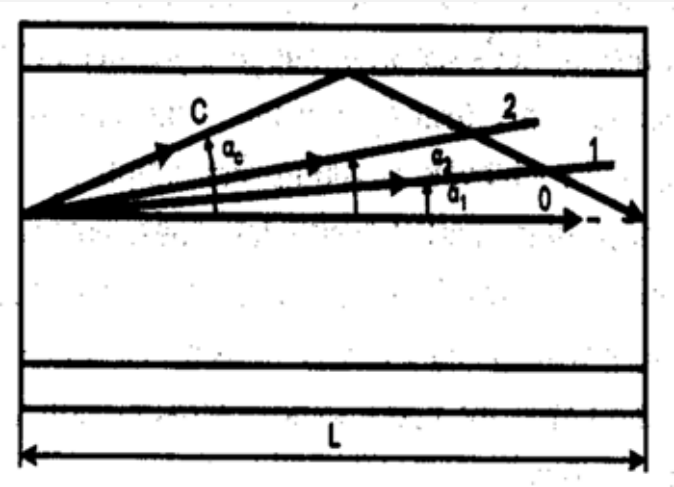
$$N = \frac{V^2}{4}$$

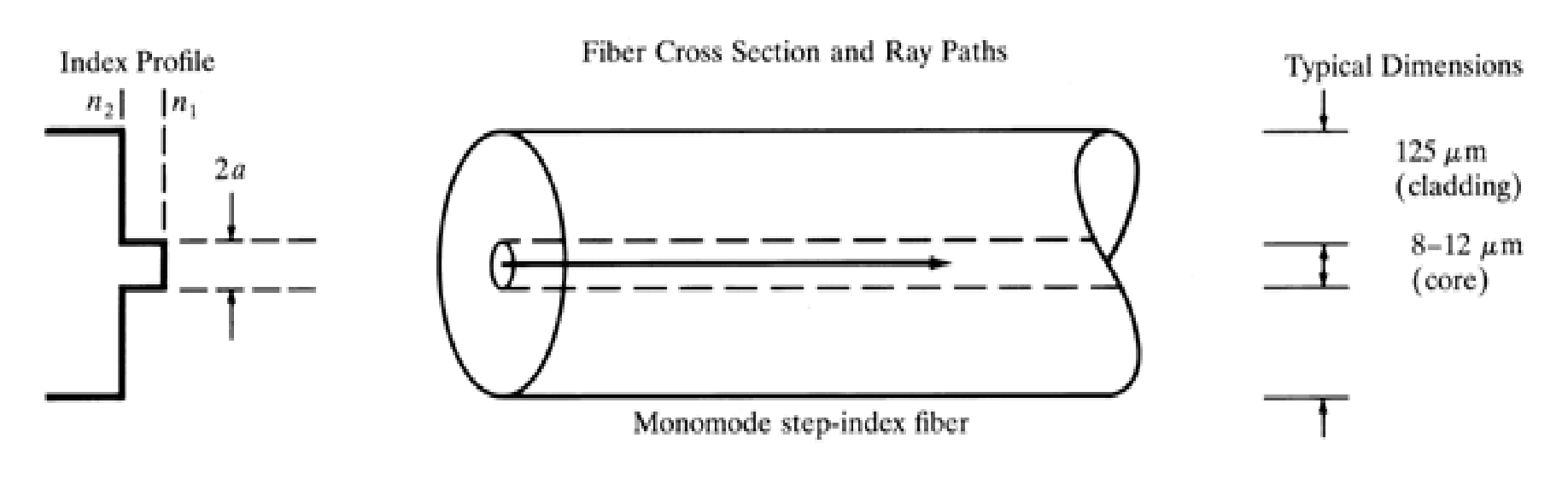
Single Mode Fiber

- The underlying reason
  - for modal dispersion problem is the existence of many modes that deliver the same light pulse
- Why not limit
  - the light beam inside the core to only one mode?
- Doing so they
  - reason would eliminate the problem of intermodal dispersion completely

Single Mode Fiber

- The result is
  - the advent of single mode fiber





Single Mode Fiber

- The principle on which single mode fiber is based is that
  - by appropriately choosing a, n_1, n_2 and λ so that

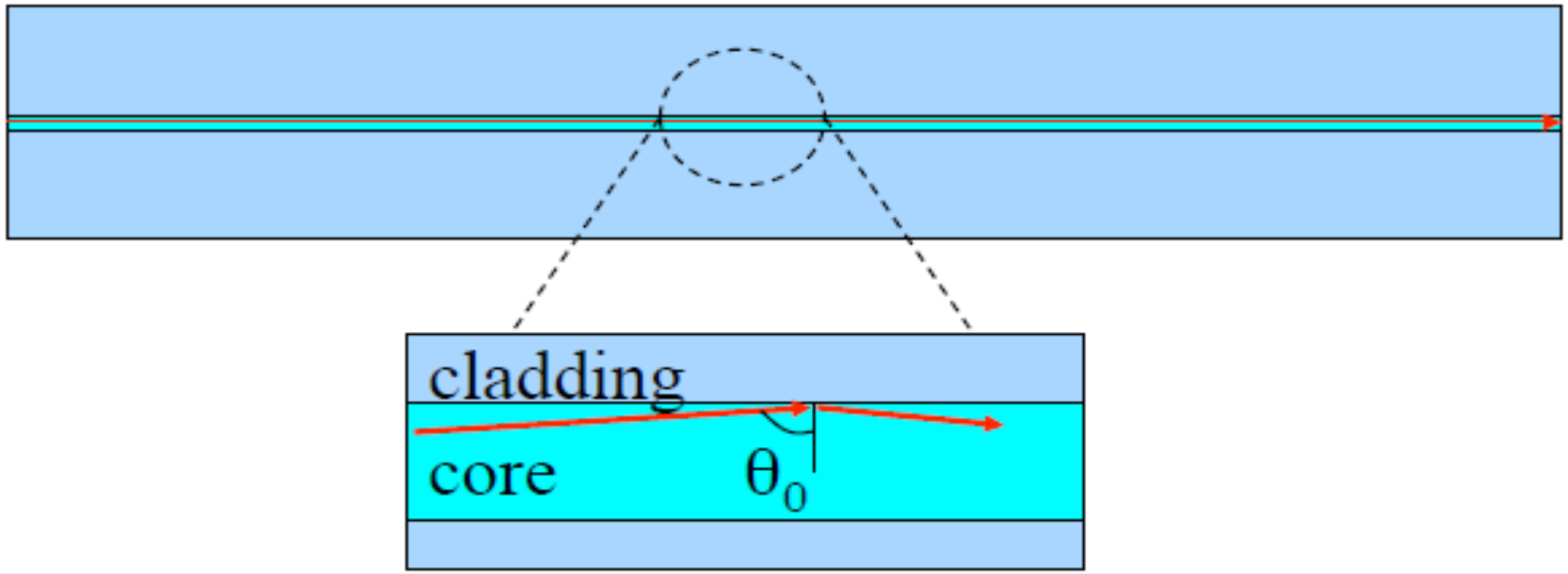
$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} \leq 2.405$$

  - all modes except the fundamental mode are cut off

- Typical
  - d and Δ are 8.3μm and 0.37% respectively

Single Mode Fiber

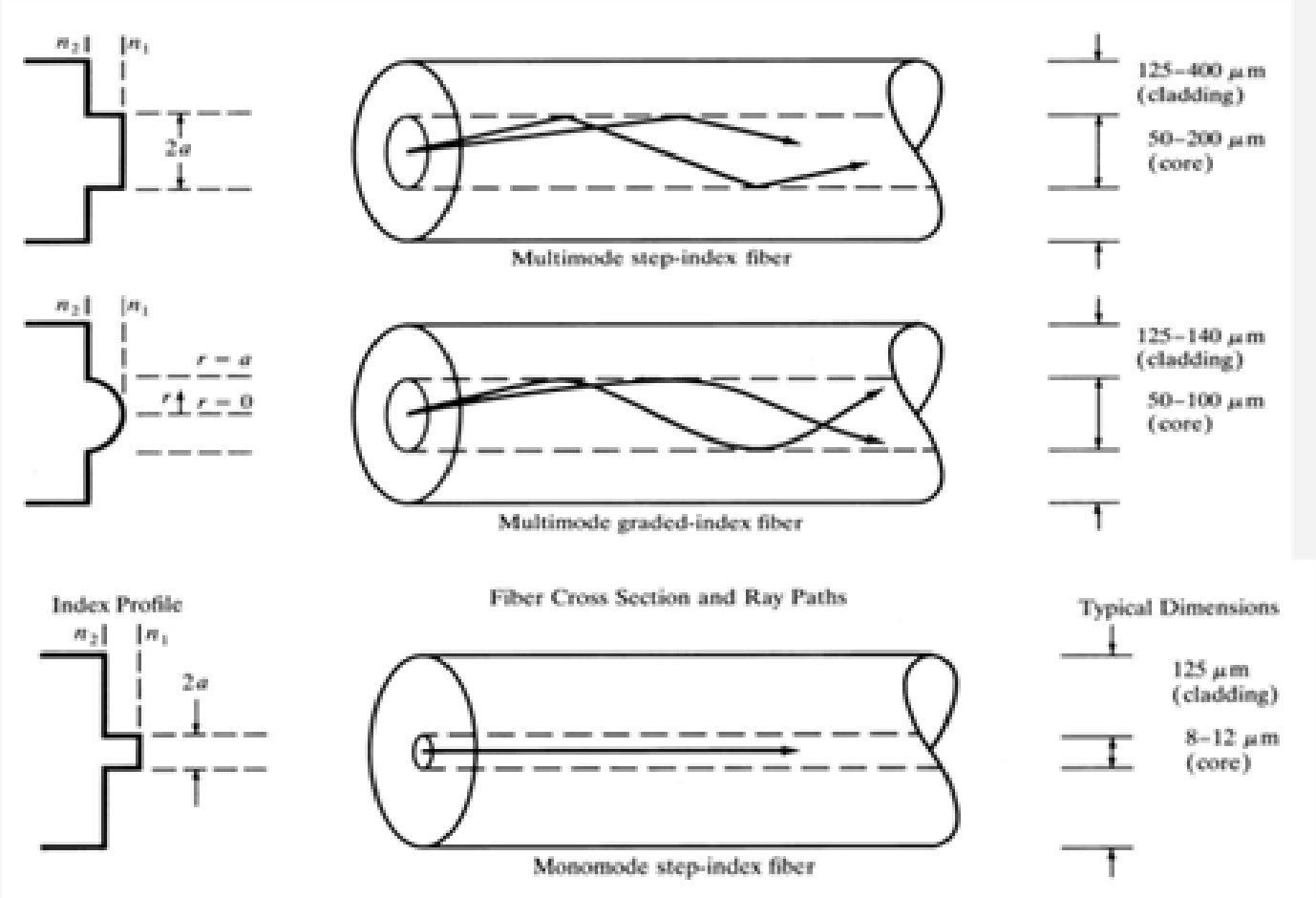
- Propagates
  - only one mode so that modal dispersion be absent



Types of Fiber

- The three basic types of fiber optic cable used in communication systems are:
  1. Step-index multimode
  2. Graded-index multimode
  3. Step-index single mode

Basic Types of Fiber

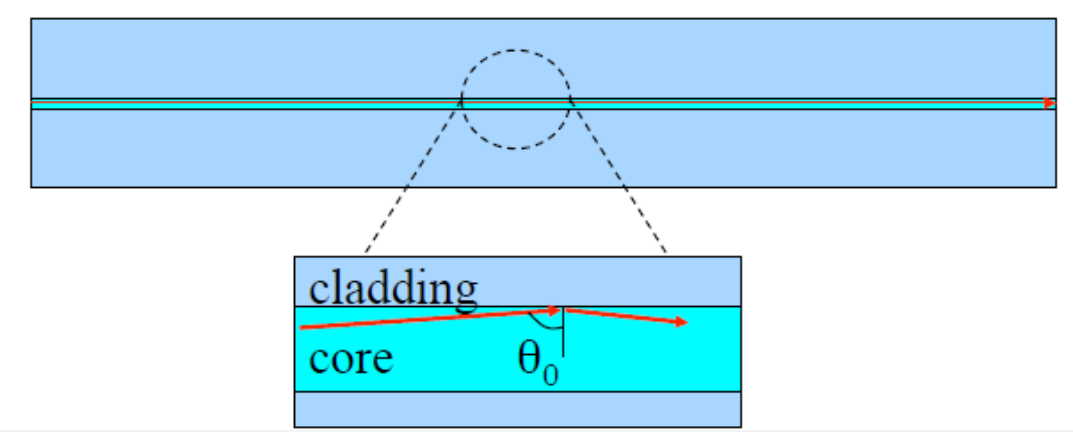


Problem

- A multimode step-index fiber has core and cladding refractive indices of 1.475 and 1.458 respectively. If it has a core radius of 30 μm and operates at a wavelength of 1250 nm.
  - What should be the core radius of the fiber if it is to operate as a single mode fiber

Single Mode Fiber (SMF)

- Propagates
  - only one mode so that modal dispersion be absent



- Pulse broadening
  - however, does not disappear altogether with the advent of SMF

Dispersion

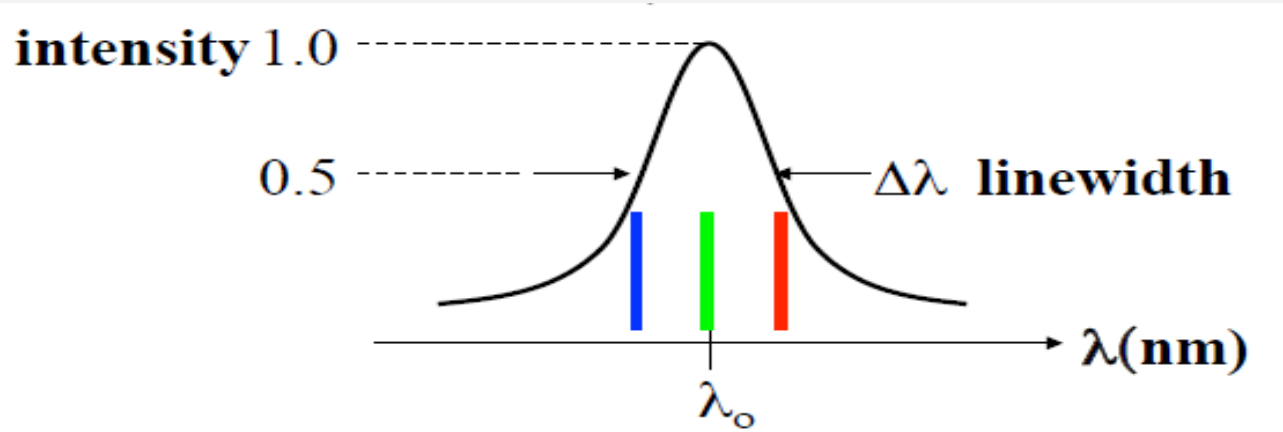
- The basic mechanism of dispersion involves
  - different light beams carrying light pulses
    - The beams arrive at the receiver end at different times, causing the output light pulses to spread

Chromatic Dispersion

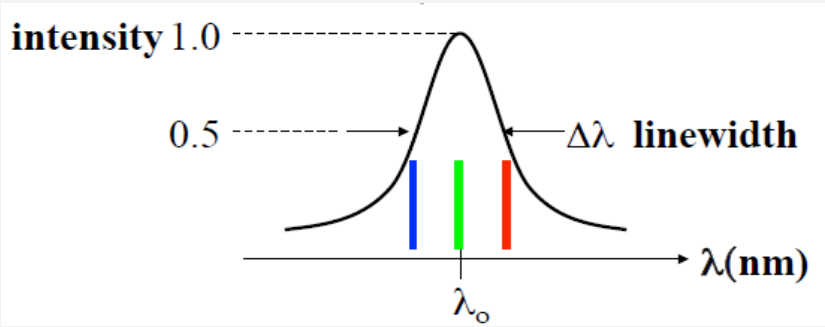
- Covers all the phenomena
  - associated with wavelength dependent pulse spreading
- It is pulse spreading due
  - to the fact that different wavelengths of light propagate at slightly different velocities through the fiber
- It arises from the
  a. variation of refractive index with wavelength
  b. dependence of the fiber's waveguide properties on wavelength

Chromatic Dispersion

- A beam is
  - composed of light having several wavelengths
    - Simply because there is no light source in nature that can radiate a single wavelength



Pulse broadening (material)

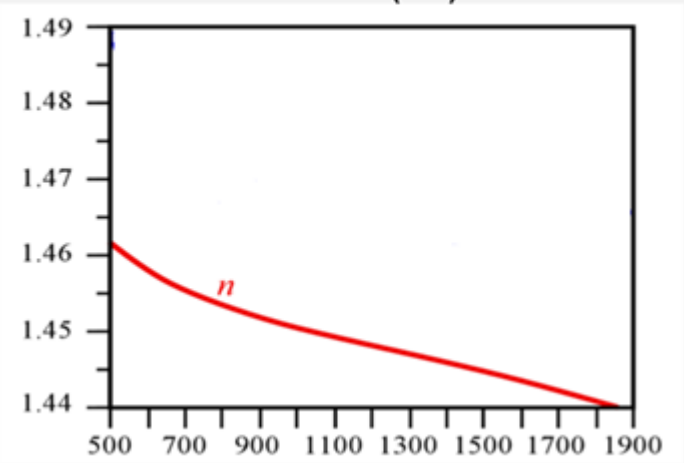


- Arises due to
  1. finite spectral width of the optical source
  2. Dependency of refractive index on wavelength

- Thus

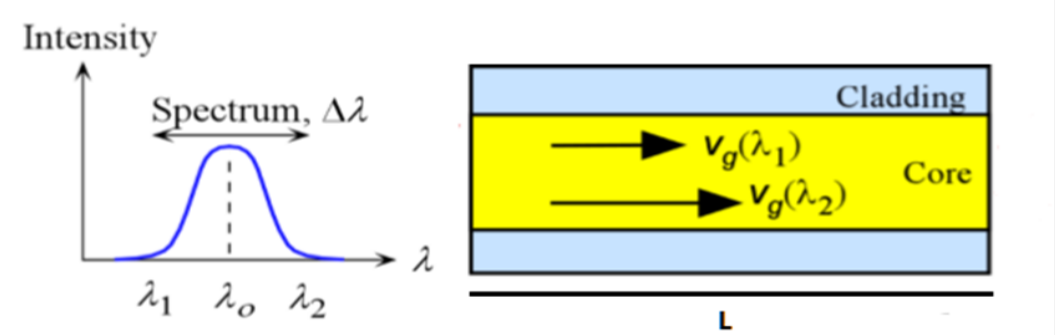
$$\upsilon = \frac{c}{n(\lambda)}$$

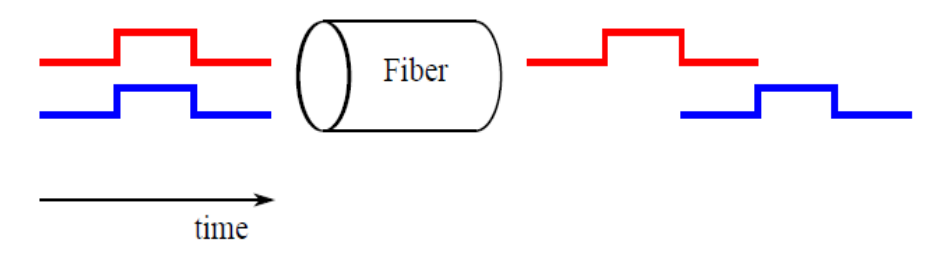
$$n = n(\lambda)$$



Pulse broadening

- Material dispersion
  - different spectral components of a pulse travel at different group velocities





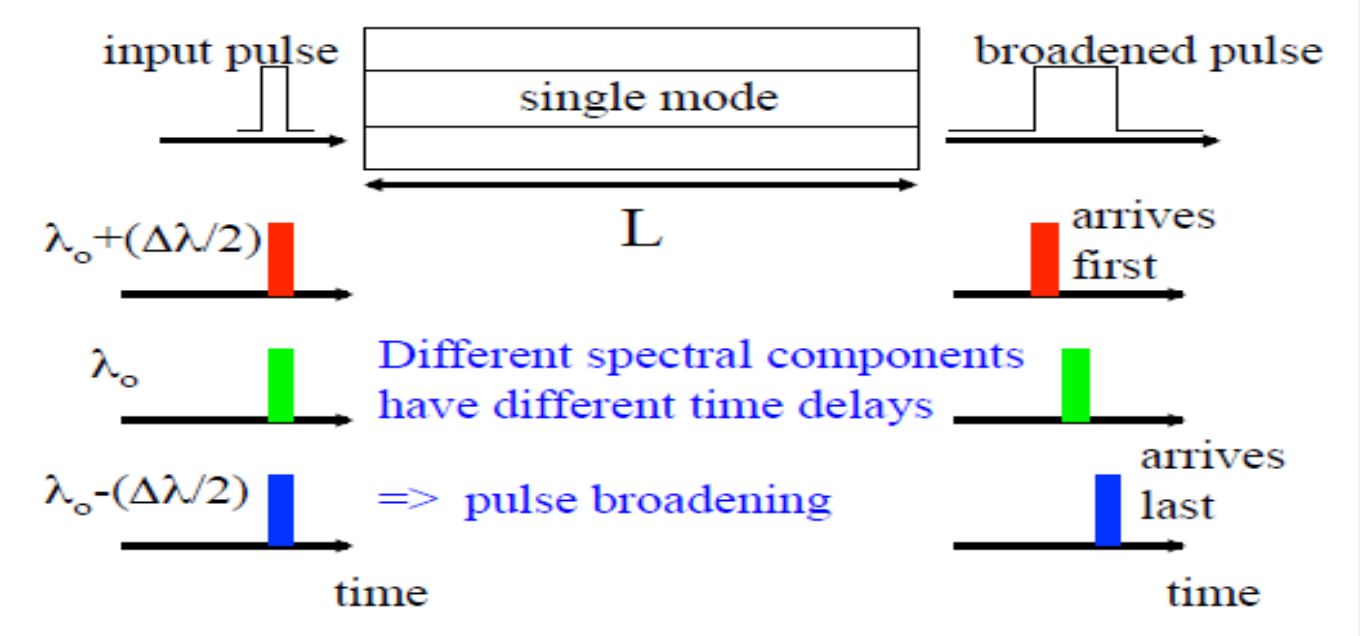
  - This is known as
    - Group Velocity Dispersion (GVD)

Chromatic Dispersion

- Light of different wavelengths travel along the fiber at
  - different velocities and will arrive at the receiver end at different times

Chromatic Dispersion

- Pulse broadening occurs
  - because there is propagation delay (group delay) differences among the spectral components of the transmitted signal



Chromatic Dispersion

- Two mechanisms are responsible for chromatic dispersion
  1. Material dispersion
  2. waveguide dispersion

$$D(\lambda) = D_{mat}(\lambda) + D_{wg}(\lambda)$$

Chromatic Dispersion

- Two mechanisms are responsible for chromatic dispersion
  1. Dependency of refractive index on wavelength n = n(λ)
     - different wavelength components travel at different speeds in silica - **Material dispersion**

$$\upsilon = \frac{c}{n(\lambda)}$$

- Thus

$$D_{mat}(\lambda) = -\frac{L}{c} \cdot \lambda \frac{d^2 n}{d\lambda^2}$$

Zero Dispersion Wavelength

- The condition for zero dispersion is when

$$\frac{d^2 n}{d\lambda^2} = 0$$

  - The value of λ for which the material dispersion is zero is called the zero- dispersion wavelength, λ_{ZD}
- For pure silica
  - the zero- dispersion wavelength, λ_{ZD} is approximately 1276 nm

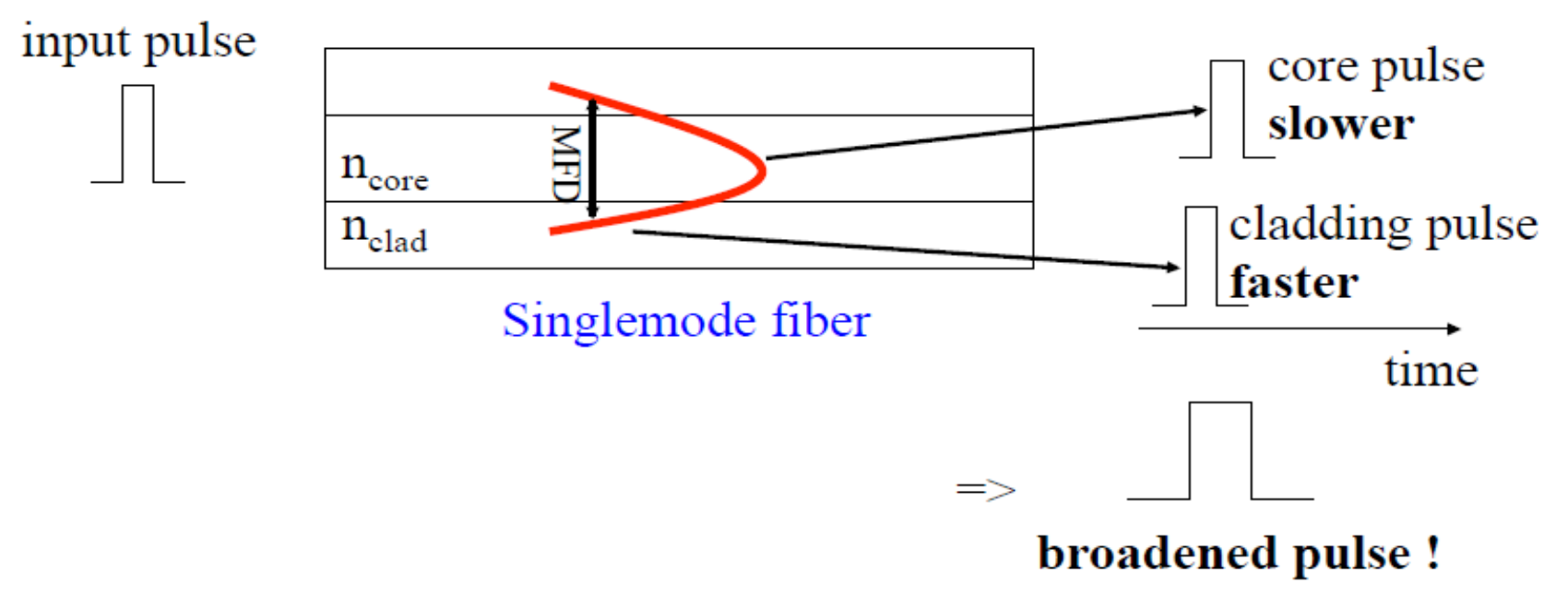
Chromatic Dispersion

2. Waveguide dispersion
   - develops when the propagation constant is dependent on wavelength

- Thus
  - Even with total internal reflection some portion of the incident field will spread into (enter) the cladding
    - Light energy of a mode propagates partly in the core and partly in the cladding
    - The mode power distribution between the core and cladding depends on wavelength (λ) – **waveguide dispersion**

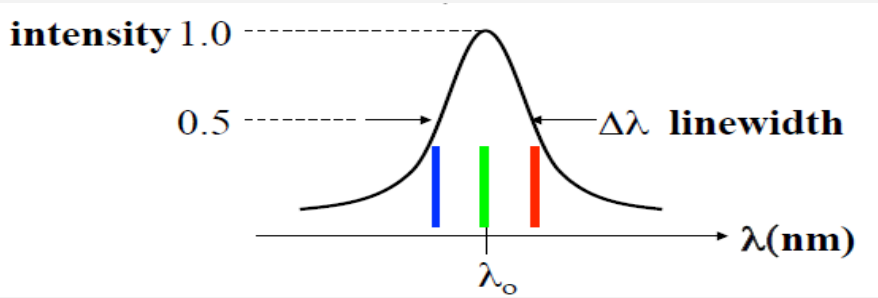
Waveguide dispersion

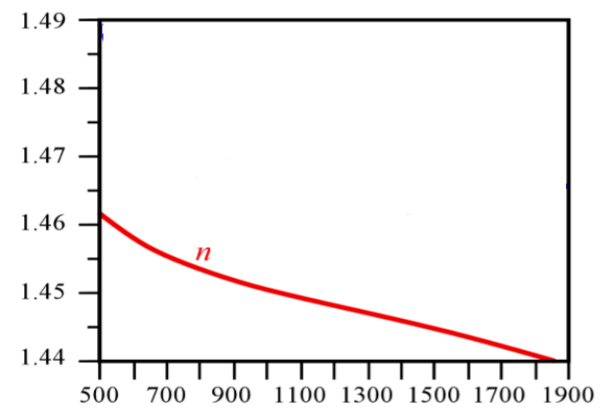
- Wave guide dispersion
  - depends on the mode field distribution in the core and cladding

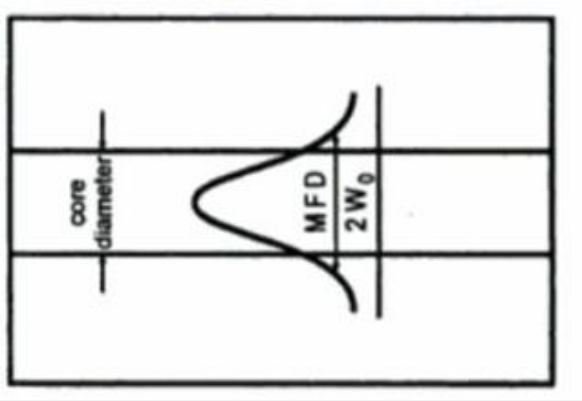


Recap: Chromatic Dispersion

- Covers all the phenomena
  - associated with wavelength dependent pulse spreading
- It arises from the
  i. finite spectral width of the optical source
  ii. variation of refractive index with wavelength
  iii. dependence of the fiber's waveguide properties on wavelength







Chromatic Dispersion

- Pulse spreading due to Chromatic dispersion is given by

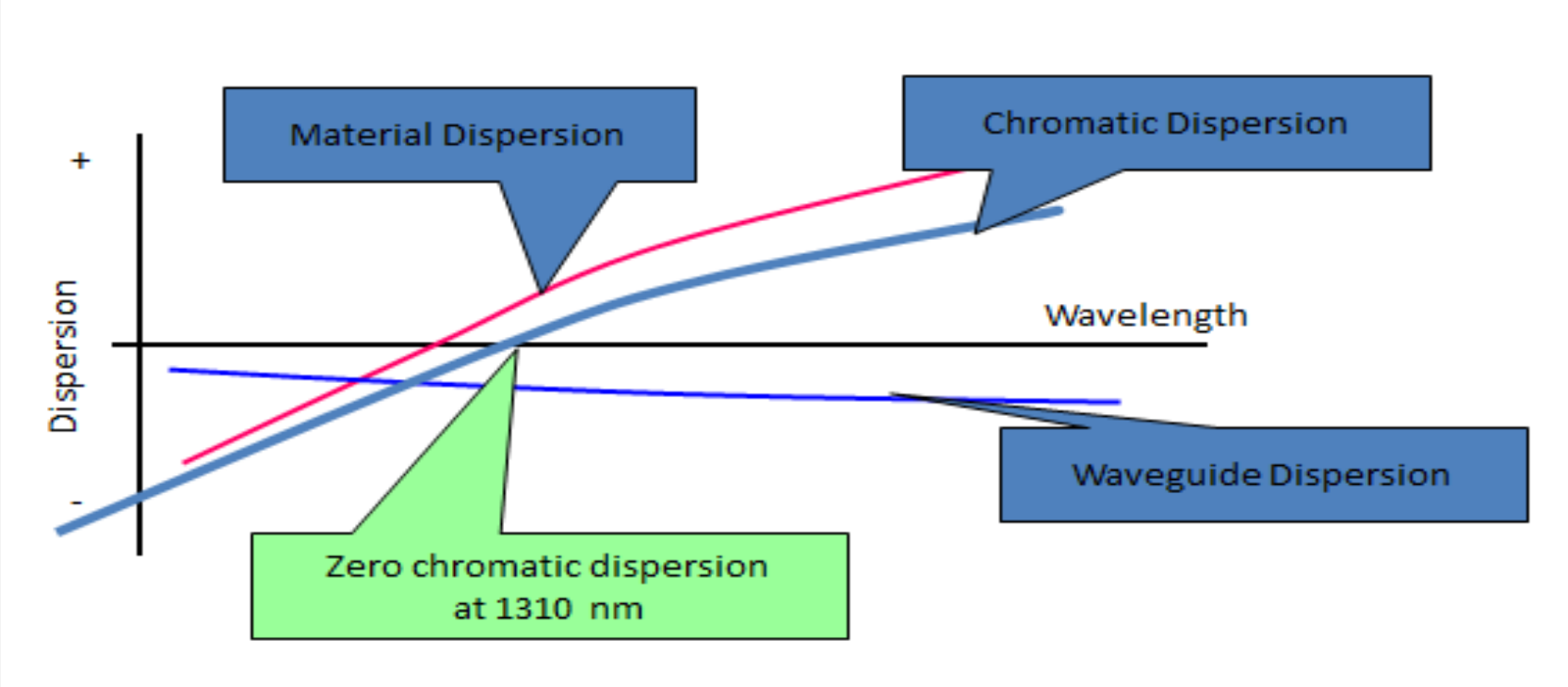
$$\Delta T = L \cdot D(\lambda) \cdot \Delta\lambda$$

- where D(λ) is called the dispersion parameter
  - Cover all the phenomena associated with wavelength dependent pulse spreading

$$D(\lambda) = D_{mat}(\lambda) + D_{wg}(\lambda)$$

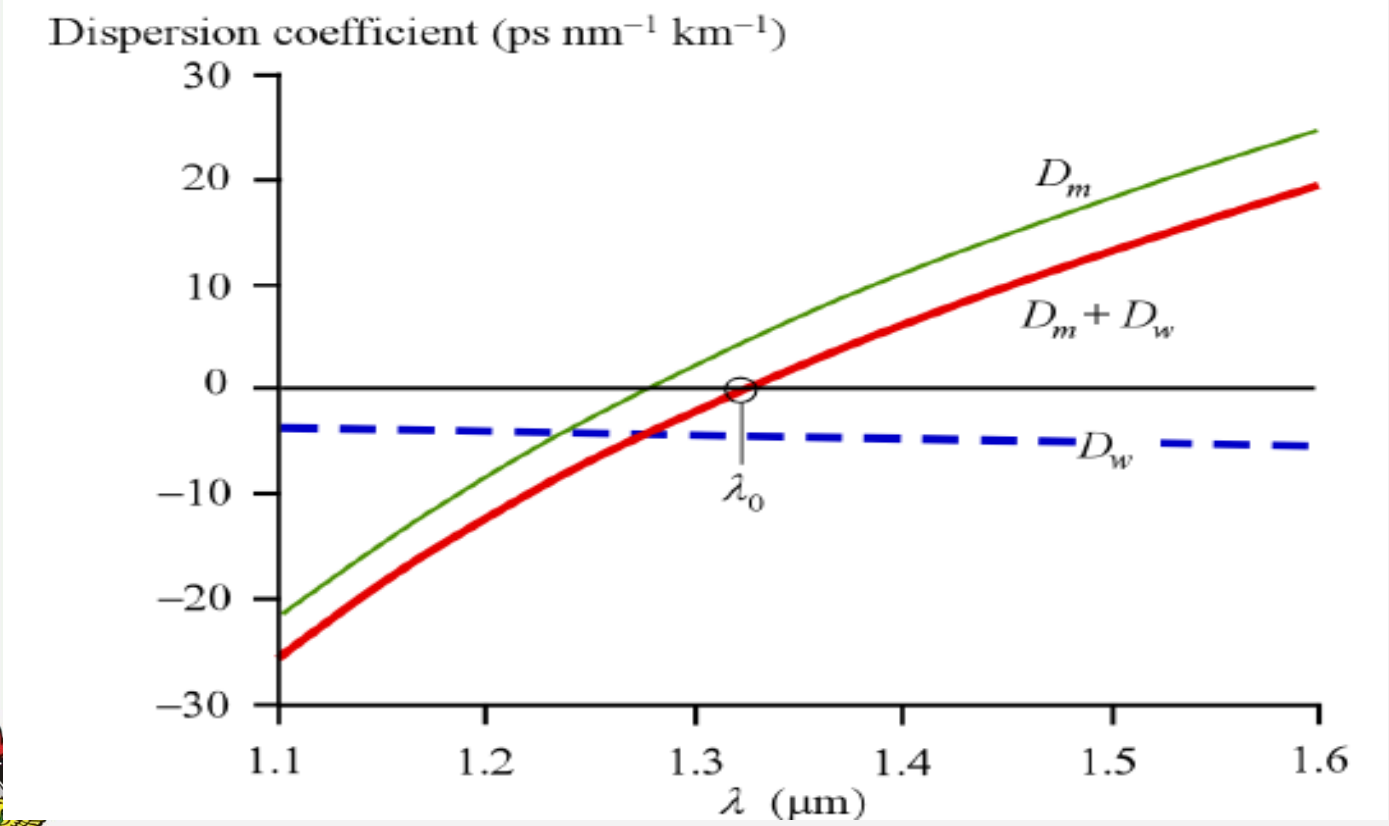
Dispersion Parameter

- Made of two parts: $D(\lambda) = D_{mat}(\lambda) + D_{wg}(\lambda)$



Chromatic Dispersion

- Chromatic dispersion is the



$$\frac{\Delta\tau}{L} = (D_m + D_w)\Delta\lambda$$

Chromatic Dispersion

- Manufacturers specify
  - the chromatic dispersion parameter for fibers either by giving its value or by using the formula

$$D(\lambda) = \frac{S_0}{4}\left[\lambda - \frac{\lambda_0^4}{\lambda^3}\right]$$

  - where
    - S_0 = the zero-dispersion slope in ps/nm^2.km
    - λ_0 = the zero-dispersion wavelength in nm
    - λ = operating wavelength in nm

Problem

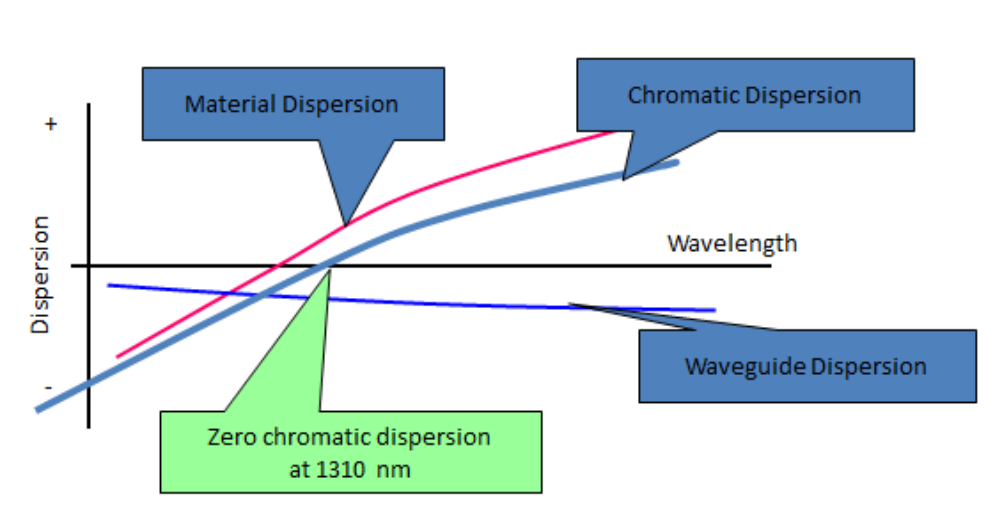
- For a 60-km single mode fiber at 1.3 and 1.55 μm with the following parameters: zero dispersion wavelength is 1310 nm, dispersion slope of 0.092 ps/(nm^2.km), and the source spectral width is 2nm.
  a. How far can optical pulses go on this fiber before overlapping?
  b. For each window estimate the limiting bit rate

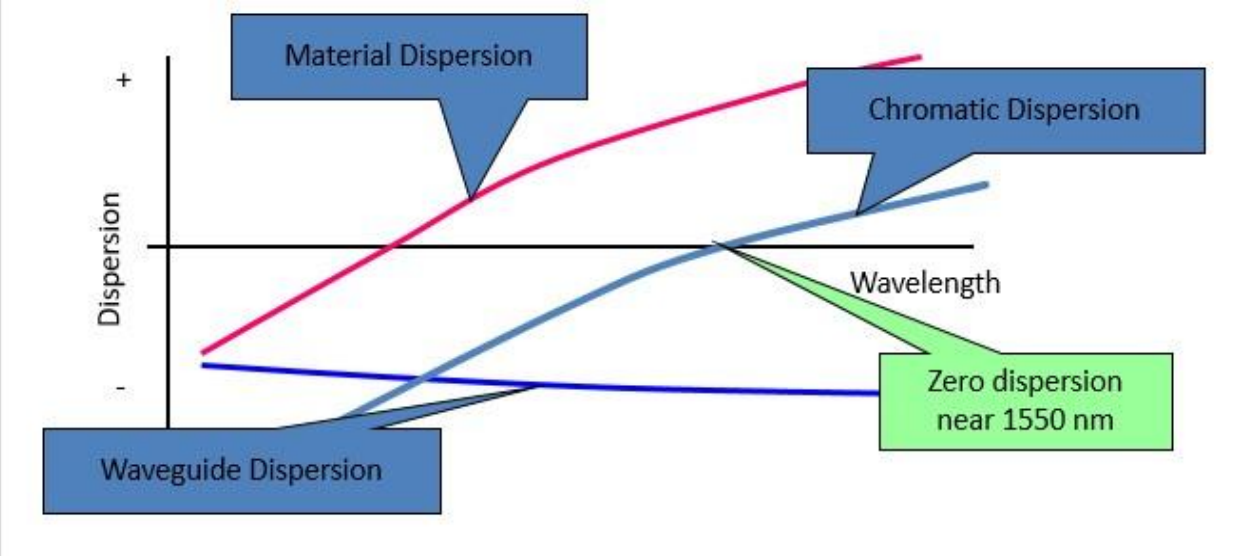
Problem

a. A manufacturer's data sheet states that a fiber has a zero-dispersion wavelength at 1312 nm, a zero-dispersion slope of 0.090 ps/nm^2.km.
   i. Compare the dispersion for this fiber at wavelengths of 1310 nm and 1550 nm.
   ii. Calculate the pulse spread caused by chromatic dispersion if the fiber has a length of 100 km and operates at 1310 nm. The spectral width is 2 nm

Dispersion shifted Fiber

- Changing waveguide dispersion shifts zero dispersion point



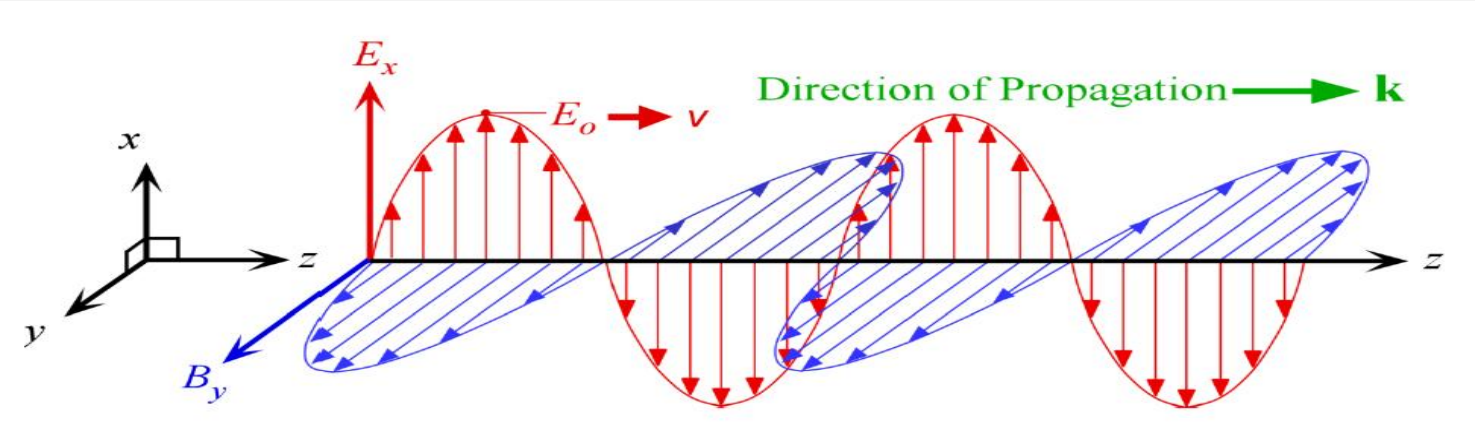


Problem

- Why is dispersion-shifted fiber required for a long-distance single mode fiber system operating at high data rates?

Polarization modes

- Light-signal energy in a single-mode fiber occupies
  - two orthogonal polarization states



PMD

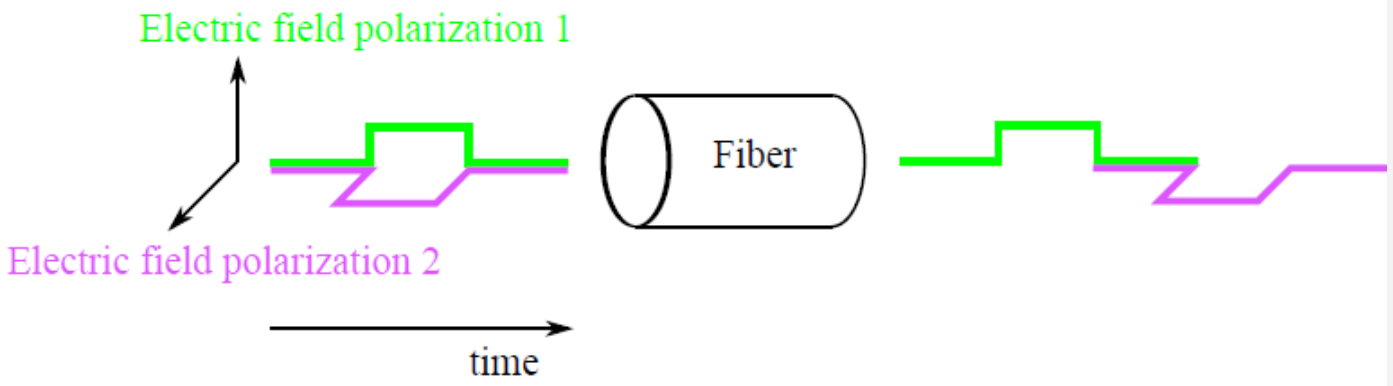
- Polarization Mode Dispersion (PMD)
  - is a form of modal dispersion in single-mode fibers where two orthogonal polarization components of a light pulse travel at
    - different speeds due to fiber birefringence, caused by imperfections or stress
      - This creates a differential group delay (DGD), leading to pulse broadening

PMD

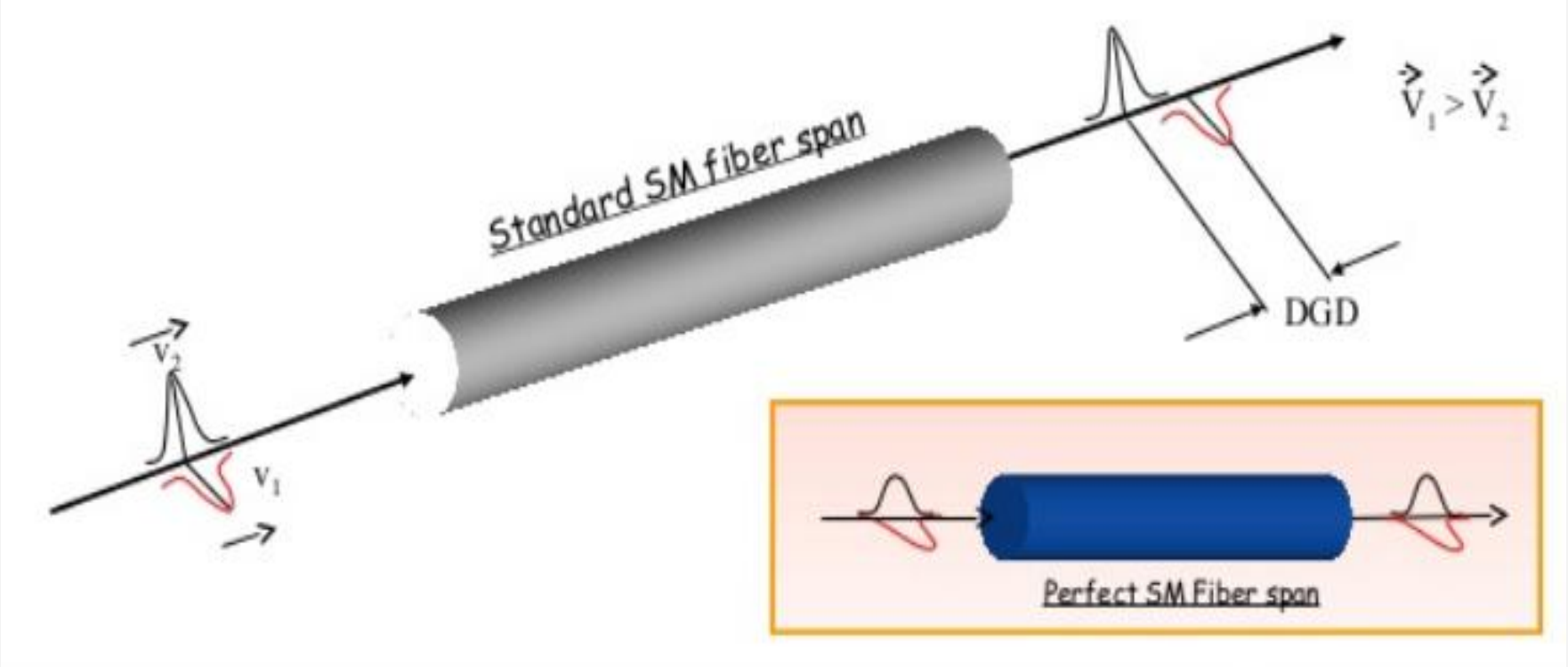
- Cause:
  - PMD arises from fiber asymmetry, such as non-circular cores (ellipticity) and mechanical or environmental stresses like temperature variations, micro-bends, or twists
- Mechanism:
  - These asymmetries create birefringence, meaning the refractive index differs for the two orthogonal polarization modes (typically horizontal and vertical), causing one to travel faster than the other

Polarization Mode Dispersion

- Due to slight variations in the fiber material, each polarization mode encounters a different refractive index and travels at a slightly different velocity
- The difference in propagation times between the two orthogonal polarization modes causes pulse spreading



Polarization Mode Dispersion (PMD)



Chromatic Dispersion

- Parameters of interest
  - Group velocity
  - Group delay
  - Group velocity dispersion parameter
  - Dispersion parameter
    - Material dispersion
      - Group refractive index
    - Waveguide dispersion
  - Zero dispersion wavelength
  - Dispersion shifted fiber

Maxwell's Theory/Equations

- To develop
  - the light propagation theory in optical fibers from a generalized point of view
    - we have to use Maxwell's theory which gives theoretical explanations to well established experimental facts on electromagnetics
- Maxwell's theory
  - is based on a set of four equations known as Maxwell's equations

Maxwell's Theory/Equations

- Like all electromagnetic phenomena
  - Propagation of optical fields in fiber is governed by Maxwell's equations
    - Maxwell's equations are key to the theoretical explanations of attenuation and dispersion
- Maxwell's equations
  - enable us to derive wave equations describing the propagation of EM waves

Maxwell's Equations

- That is

$$\nabla^2 E = -\mu\varepsilon \frac{\partial^2 E}{\partial t^2}$$

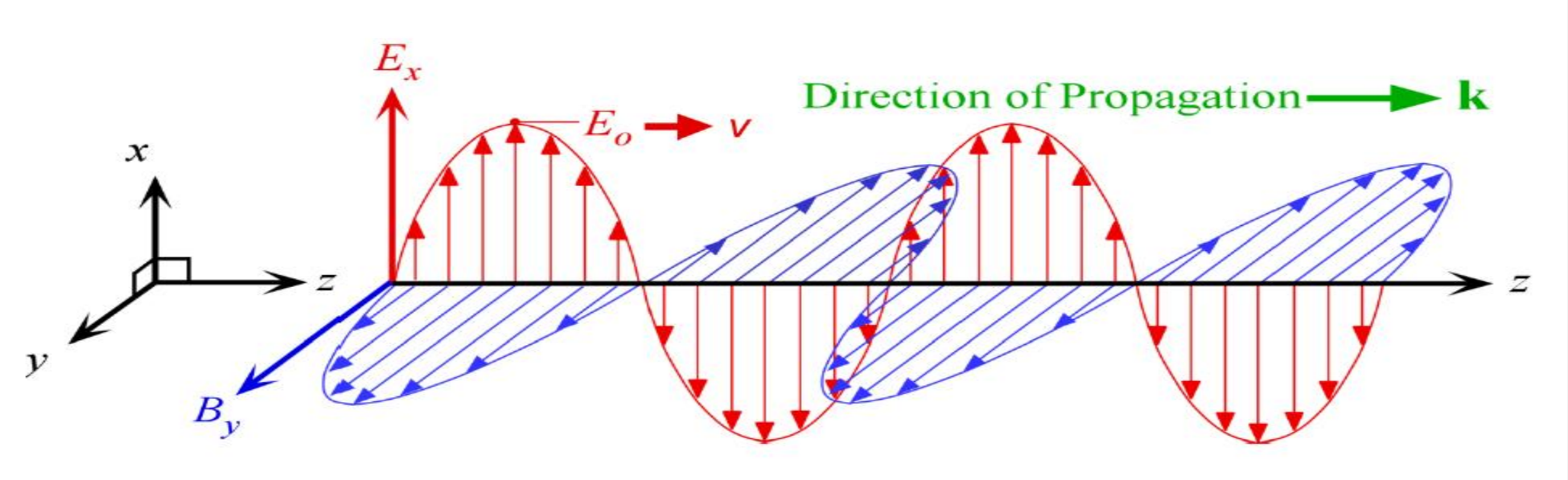
- This is second order differential equation and the solution is

$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z)$$

and

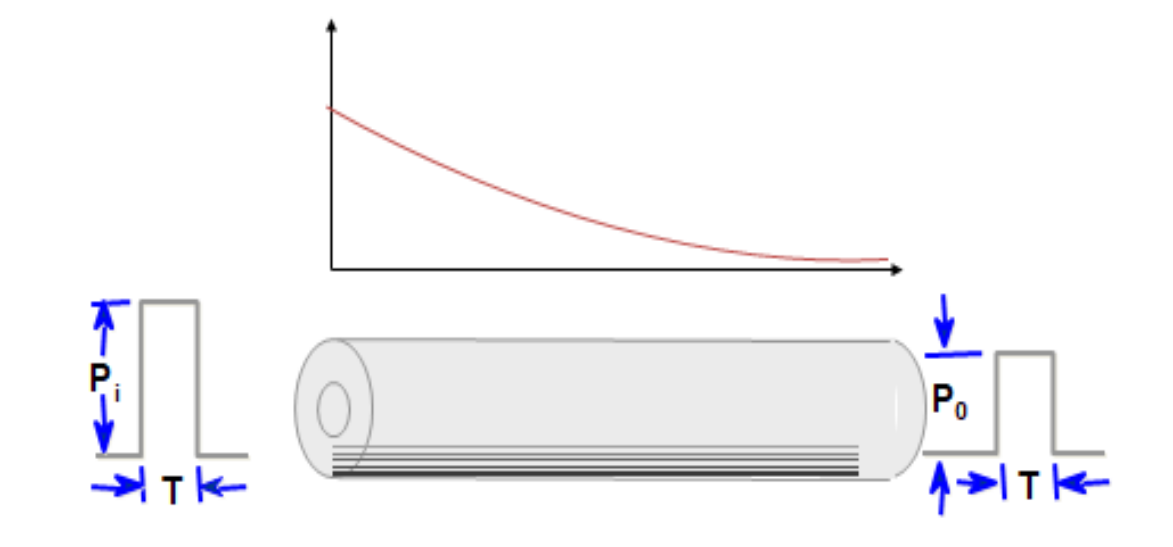
$$c = \frac{1}{\sqrt{\frac{\varepsilon}{\mu}}}$$

Wave View of Light



$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z)$$

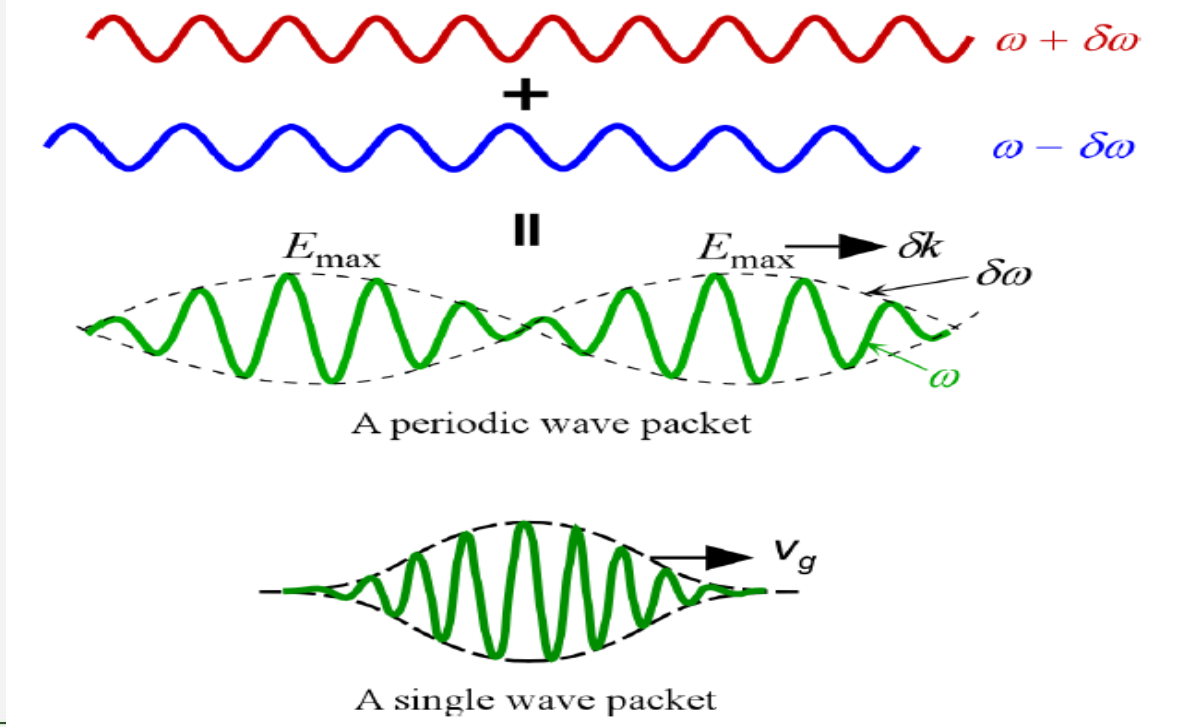
Optical Attenuation



Attenuation in dB
Power is measured in dBm:

Group Velocity - Concept

- A group of waves
  - differing slightly in wavelength combine to carry information
    - Two slightly different wavelength waves travelling in the same direction result in a wave packet
  - Group Velocity
    - is the velocity at which the energy or information is conveyed along a wave



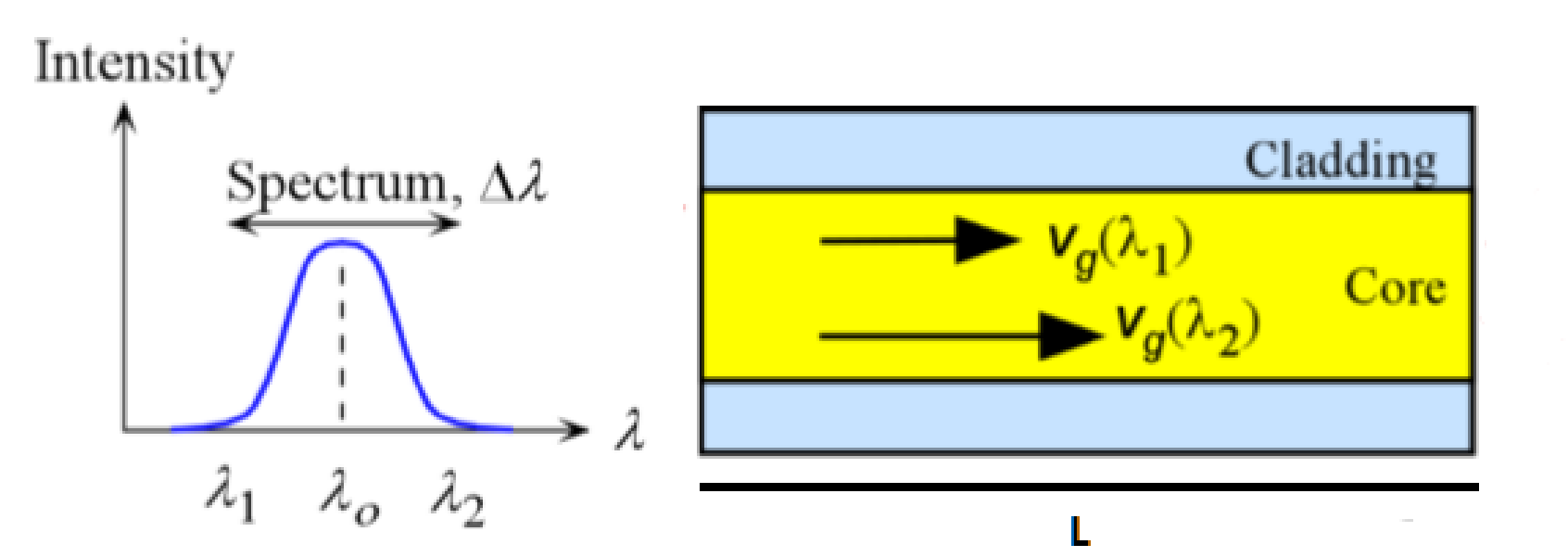
Group Velocity

- It is the velocity of the envelop and is given by

$$V_g = \frac{\partial \omega}{\partial \beta} = \left(\frac{\partial \beta}{\partial \omega} \right)^{-1}$$

Group-Velocity Dispersion(GVD)

- Consider a single mode fiber of length L



Group-Velocity Dispersion(GVD)

- As the signal propagates along the fiber
  - a specific spectral component at a frequency ω (wavelength λ) would arrive at the end of the fiber a time delay T (group delay)
  - The group delay is

$$T = \frac{L}{V_g}$$

Group-Velocity Dispersion(GVD)

- The parameter

$$\frac{d^2\beta}{d\omega^2} = \beta_2$$

  - is called the Group-Velocity Dispersion (GVD) parameter
  - shows that the group velocity depends on frequency
    - This parameter determines how much an optical pulse would broaden on propagating inside the fiber
- Thus the pulse spreading due to differential time delay is

$$\Delta T = L \cdot \beta_2 \cdot \Delta\omega$$

Dispersion Parameter

- Thus

$$\Delta T = L \cdot D(\lambda) \cdot \Delta\lambda$$

  - where D(λ) is called the dispersion parameter
    - Cover all the phenomena associated with wavelength dependent pulse spreading

- Pulse spreading as a function of wavelength is

$$D(\lambda) = \frac{-2\pi c}{\lambda^2}\beta_2$$

Maxwell's Equations

- Electromagnetic radiation propagates in the form of two mutually coupled vector waves
  - Electric field wave
  - Magnetic field wave
- Maxwell's equations in differential form is

$$\nabla \cdot D = \rho_{\upsilon} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$\nabla \cdot B = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

$$\nabla \times H = J + \frac{\partial D}{\partial t} \ldots\ldots\ldots\ldots\ldots (4)$$

Maxwell's Equations

- Where

$$D = \varepsilon E$$

$$B = \mu H$$

$$J = \sigma E$$

  - E= the electric field intensity
  - D= the electric flux density
  - J= the current density
  - H= the magnetic field intensity
  - B= the magnetic flux density
  - ρ_v = the volume charge density

Maxwell's Equations

- Maxwell's equations
  1. Describe the sources of electromagnetic fields
  2. Enable us to derive wave equations describing the propagation of EM waves

- Interpretation of Maxwell equations
  - Eqn1
    - That the flow source of electric field is a free charge
      - There is a free electric charge in nature
  - Eqn2
    - There is no flow source for magnetic field
      - There is no free magnetic charge in nature

Maxwell's Equations

- Interpretation of Maxwell equations
  - Eqn3
    - That the rotating nature of an electric field is as a result of time varying magnetic field
  - Eqn4
    - That the rotating nature of a time varying magnetic field is the time changing density of conduction and displacement

Maxwell's Equations

- That is

$$\nabla^2 E = -\mu\varepsilon \frac{\partial^2 E}{\partial t^2} \ldots\ldots\ldots\ldots\ldots\ldots\ldots(7)$$

- This is second order differential equation and the solution is

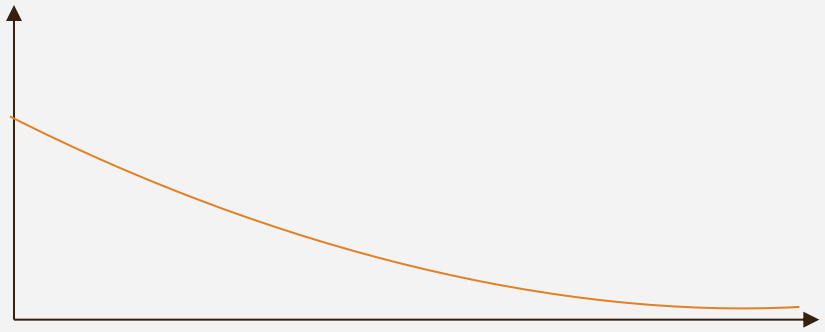
$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z)$$

- and

$$c = \frac{1}{\sqrt{\frac{\varepsilon}{\mu}}}$$

Optical Attenuation

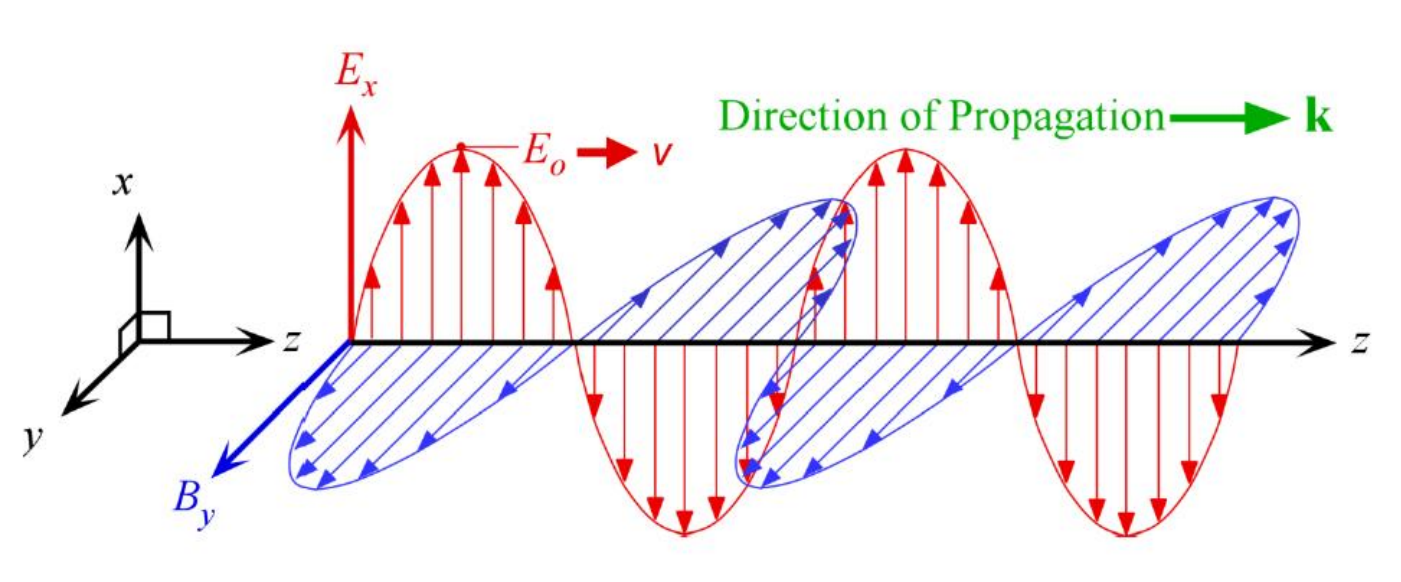
- Attenuation in dB
- Power is measured in dBm:



Pulse Spreading due to Chromatic Dispersion

- From the wave view of light
  - The z-polarized component of an electric wave takes the general form

$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z)$$



Pulse Spreading due to Chromatic Dispersion

- ω = angular frequency
  - Scalar measure of rotation rate
- β = propagation constant
  - Measure of the change undergone by the amplitude of the wave as it propagates in a given direction
- Lets take a point M
  - on the wave and determine at what velocity this point moves
- Point M is
  - constant with respect to the wave i.e

$$\omega t - \beta z = cons\tan t$$

$$\omega t - \beta z = k$$

Pulse Spreading due to Chromatic Dispersion

- Thus

$$\beta z = \omega t - k$$

$$z = \frac{\omega}{\beta} t - \frac{k}{\beta}$$

- The velocity (phase velocity) at which M moves is given by

$$\upsilon = \frac{dz}{dt} = \frac{\omega}{\beta}$$

Pulse Spreading due to Chromatic Dispersion

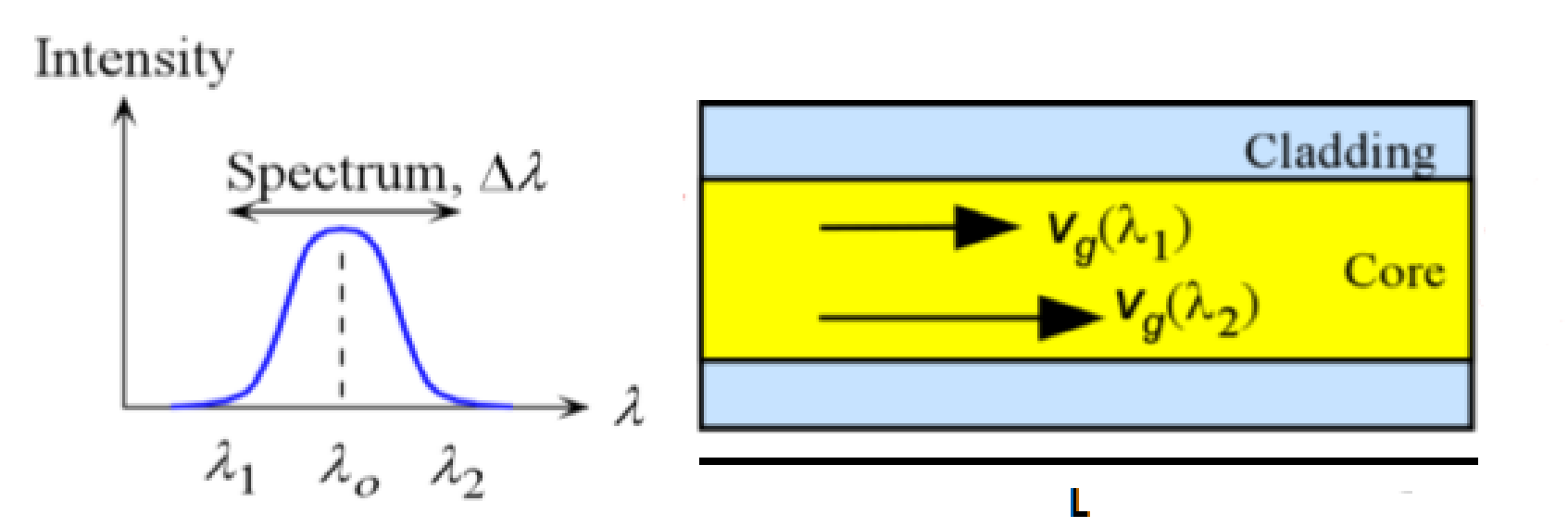
- Group velocity
  - is the velocity of the envelop and is given by

$$V_g = \frac{\partial \omega}{\partial \beta} = \left(\frac{\partial \beta}{\partial \omega} \right)^{-1}$$

- the group velocity
  - is the velocity at which the energy in a pulse travels along a fiber
  - Therefore defines the speed with which energy or information is propagated
  - depends on frequency/wavelength and the different group velocities of the frequency/wavelength components of a pulse cause it to broaden as it travels along a fiber
  - of a wave is the velocity with which the overall shape of the wave's amplitude propagates

Group-Velocity Dispersion(GVD)

- Consider a single mode fiber of length L



- As the signal propagates along the fiber
  - a specific spectral component at a frequency ω (wavelength λ) would arrive at the end of the fiber a time delay T (group delay)

Group-Velocity Dispersion(GVD)

- The group delay is

$$T = \frac{L}{V_g}$$

- The quantity of interest is
  - the amount of pulse spreading that arises from the group delay variation
- The delay difference
  - per unit frequency along the propagation path is approximately

$$\frac{dT}{d\omega}$$

- For spectral components that are Δω apart

Group-Velocity Dispersion(GVD)

- The total delay difference ΔT over a fiber of length L is

$$\Delta T = \frac{dT}{d\omega} \cdot \Delta\omega$$

- But

$$T = \frac{L}{V_g}$$

- That is

$$\Delta T = \frac{d\left(\frac{L}{V_g}\right)}{d\omega} \cdot \Delta\omega$$

Group-Velocity Dispersion(GVD)

- But

$$\frac{1}{V_g} = \frac{\partial \beta}{\partial \omega}$$

- Total delay difference is therefore

$$\Delta T = L \cdot \frac{d\left(\frac{d\beta}{d\omega}\right)}{d\omega} \cdot \Delta\omega$$

$$\Delta T = L \cdot \frac{d^2\beta}{d\omega^2} \cdot \Delta\omega$$

Group-Velocity Dispersion(GVD)

- The parameter

$$\frac{d^2\beta}{d\omega^2} = \beta_2$$

  - is called the Group-Velocity Dispersion (GVD) parameter
  - shows that the group velocity depends on frequency
    - This parameter determines how much an optical pulse would broaden on propagating inside the fiber
- Thus the pulse spreading due to differential time delay is

$$\Delta T = L \cdot \beta_2 \cdot \Delta\omega$$

Group-Velocity Dispersion(GVD)

- Changing from frequency to wavelength we use

$$\omega = \frac{2\pi c}{\lambda} = 2\pi c \lambda^{-1}$$

$$\frac{\Delta\omega}{\Delta\lambda} = -2\pi c \lambda^{-2}$$

$$\Delta\omega = -\frac{2\pi c}{\lambda^2} \cdot \Delta\lambda$$

$$\Delta T = L \cdot \left(\frac{-2\pi c}{\lambda^2} \right) \cdot \beta_2 \cdot \Delta\lambda$$

Dispersion Parameter

- Thus

$$\Delta T = L \cdot D(\lambda) \cdot \Delta\lambda$$

  - where D(λ) is called the dispersion parameter
    - Cover all the phenomena associated with wavelength dependent pulse spreading
- Pulse spreading as a function of wavelength is

$$D(\lambda) = \frac{-2\pi c}{\lambda^2}\beta_2$$

- Made of two parts

$$D(\lambda) = D_{mat}(\lambda) + D_{wg}(\lambda)$$

Material Dispersion

- Material dispersion
  - is pulse broadening due to the fact that the index of refraction varies as a function of wavelength
- We need to
  - determine the relation between material dispersion, wavelength and group refractive index
- Material dispersion
  - is the derivative of group delay with respect to wavelength

$$D_{mat}(\lambda) = \frac{dT}{d\lambda} = \frac{d\left(\frac{L}{V_g}\right)}{d\lambda} = L\frac{d\left(\frac{d\beta}{d\omega}\right)}{d\lambda}$$

  - We need to express $\frac{d\beta}{d\omega}$ in λ

Material Dispersion

- We know that

$$\beta = \frac{\omega}{\upsilon} = \frac{\omega n(\omega)}{c}$$

- thus

$$\frac{d\beta}{d\omega} = \frac{1}{c}\left[\frac{d\left(\omega \cdot n(\omega)\right)}{d\omega}\right]$$

$$\frac{d\beta}{d\omega} = \frac{1}{c}\left[n + \frac{\omega dn}{d\omega}\right] \ldots\ldots\ldots\ldots\ldots(1)$$

Material Dispersion

- But

$$\frac{dn}{d\omega} = \frac{dn}{d\lambda} \cdot \frac{d\lambda}{d\omega}$$

$$\omega = 2\pi c \lambda^{-1}$$

$$\frac{d\omega}{d\lambda} = -\frac{2\pi c}{\lambda^2}$$

- That is

$$\frac{d\lambda}{d\omega} = -\frac{\lambda^2}{2\pi c}$$

Material Dispersion

- Thus

$$\frac{dn}{d\omega} = -\frac{\lambda^2}{2\pi c} \cdot \frac{dn}{d\lambda} \text{.....(2)}$$

- Substituting (2) into (1) we have

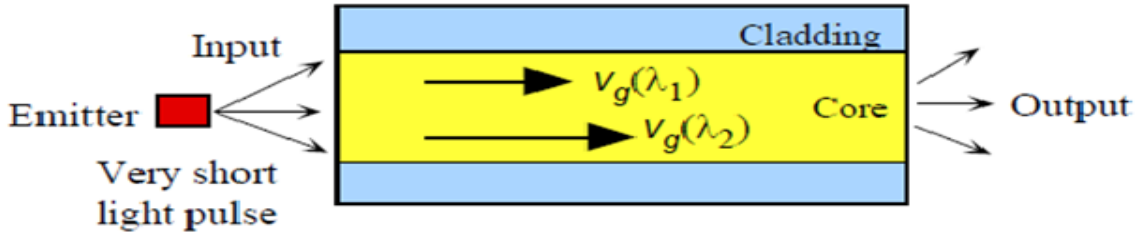
$$\frac{d\beta}{d\omega} = \frac{1}{c}\left[n - \lambda \frac{dn}{d\lambda}\right]$$

- The group refractive index, n_g, is

$$n_g = \left[n - \lambda \frac{dn}{d\lambda}\right]$$

Material Dispersion

- The group refractive index
  - determines the effect of the medium on the group velocity



$$v_g = c / n_g$$

Group velocity

Depends on the wavelength

$$T = \frac{L}{V_g} = \frac{L}{c} n_g$$

Material Dispersion

- For a distance of L, the group delay is

$$T = \frac{L}{c}\left[n - \lambda \frac{dn}{d\lambda}\right]$$

- The material dispersion is thus

$$D_{mat}(\lambda) = \frac{L}{c} \cdot \frac{d\left[n - \lambda \cdot \frac{dn}{d\lambda}\right]}{d\lambda}$$

Material Dispersion

- That is

$$D_{mat}(\lambda) = \frac{L}{c} \cdot \frac{dn_g}{d\lambda}$$

  - Material dispersion is related to the wavelength, λ, and the group refractive index, n_g
- Or from

$$D_{mat}(\lambda) = \frac{L}{c} \cdot \frac{d\left[n(\lambda) - \lambda \cdot \frac{dn}{d\lambda}\right]}{d\lambda}$$

Material Dispersion

- That is

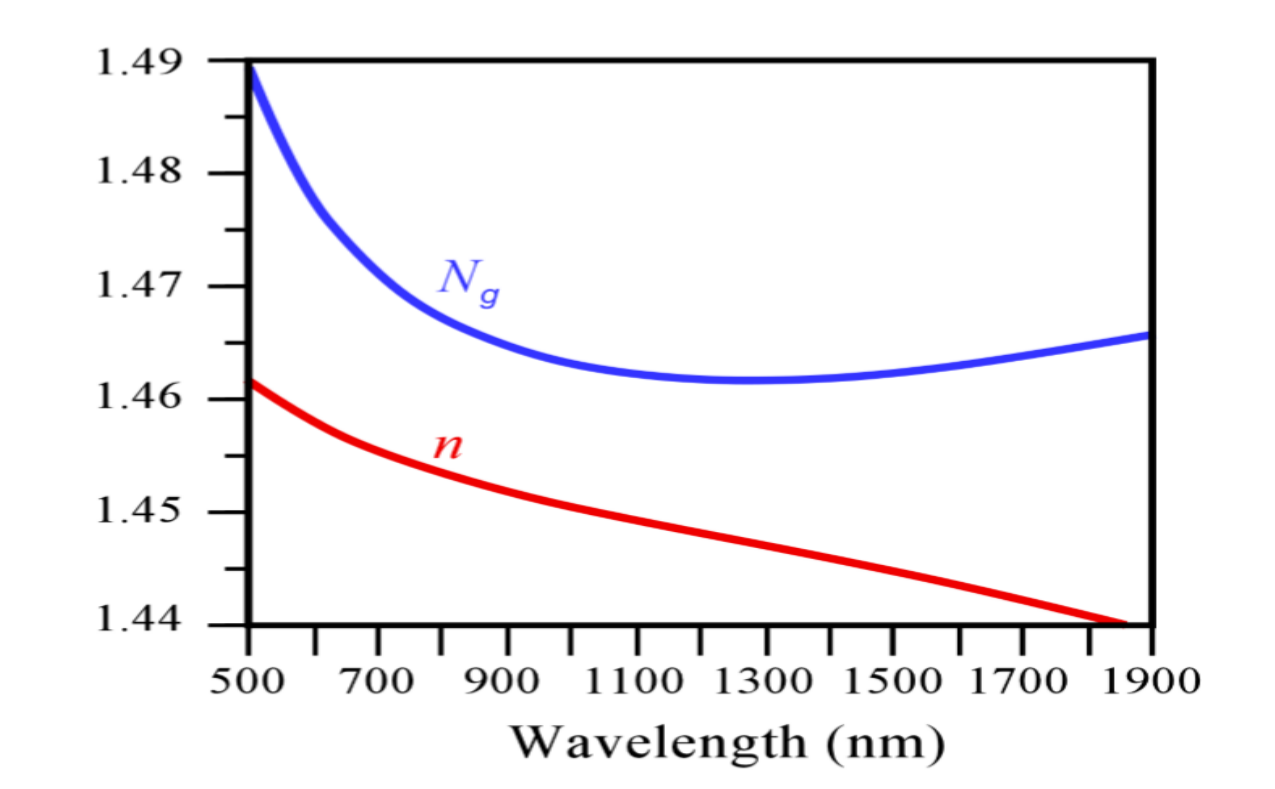
$$D_{mat}(\lambda) = \frac{L}{c} \cdot \left[\frac{dn}{d\lambda} - \lambda \frac{d^2 n}{d\lambda^2} - \frac{dn}{d\lambda} \right]$$

- This

$$D_{mat}(\lambda) = -\frac{L}{c} \cdot \lambda \frac{d^2 n}{d\lambda^2}$$

Group refractive index vs. wavelength

- N_g vs. λ



Problem

- If the group velocity of an optical pulse over a fiber of length *L* is v_g, then the group delay is given by

$$\tau = \frac{L}{v_g}$$

  - Show that the material dispersion coefficient can be approximated as

$$D_m = \frac{1}{L}\frac{d\tau}{d\lambda} \approx -\frac{\lambda}{c}\left(\frac{d^2 n}{d\lambda^2}\right)$$

Problem

- Given that $\left|\frac{d^2n}{d\lambda^2}\right|$ is 4.0 x 10^{-2} μm^{-2}.
  - Calculate the pulse broadening per km due to material dispersion within a fiber when it is illuminated with an LED with a peak wavelength of 0.9 μm and spectral width of 45 nm

THANK YOU

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